

Laws of Large Numbers

2.1 Independence

Measure theory ends and probability begins with the definition of independence. We begin with what is hopefully a familiar definition and then work our way up to a definition that is appropriate for our current setting.

Two events A and B are **independent** if $P(A \cap B) = P(A)P(B)$.

Two random variables X and Y are **independent** if for all $C, D \in \mathcal{R}$,

$$P(X \in C, Y \in D) = P(X \in C)P(Y \in D)$$

i.e., the events $A = \{X \in C\}$ and $B = \{Y \in D\}$ are independent.

Two σ -fields $\mathcal{F}$ and $\mathcal{G}$ are **independent** if for all $A \in \mathcal{F}$ and $B \in \mathcal{G}$ the events A and B are independent.

As the next result shows, the second definition is a special case of the third.

Theorem 2.1.1 (i) If X and Y are independent then $\sigma(X)$ and $\sigma(Y)$ are. (ii) Conversely, if $\mathcal{F}$ and $\mathcal{G}$ are independent, $X \in \mathcal{F}$, and $Y \in \mathcal{G}$, then X and Y are independent.

Proof (i) If $A \in \sigma(X)$, then it follows from the definition of $\sigma(X)$ that $A = \{X \in C\}$ for some $C \in \mathcal{R}$. Likewise, if $B \in \sigma(Y)$, then $B = \{Y \in D\}$ for some $D \in \mathcal{R}$, so using these facts and the independence of X and Y ,

$$P(A \cap B) = P(X \in C, Y \in D) = P(X \in C)P(Y \in D) = P(A)P(B)$$

(ii) Conversely, if $X \in \mathcal{F}$, $Y \in \mathcal{G}$, and $C, D \in \mathcal{R}$, it follows from the definition of measurability that $\{X \in C\} \in \mathcal{F}$, $\{Y \in D\} \in \mathcal{G}$. Since $\mathcal{F}$ and $\mathcal{G}$ are independent, it follows that $P(X \in C, Y \in D) = P(X \in C)P(Y \in D)$. $\square$

The first definition is, in turn, a special case of the second.

Theorem 2.1.2 (i) If A and B are independent, then so are A^c and B , A and B^c , and A^c and B^c . (ii) Conversely events A and B are independent if and only if their indicator random variables 1_A and 1_B are independent.

Proof (i) Subtracting $P(A \cap B) = P(A)P(B)$ from $P(B) = P(B)$ shows $P(A^c \cap B) = P(A^c)P(B)$. The second and third conclusions follow by applying the first one to the pairs of independent events (B, A) and (A, B^c) .

(ii) If $C, D \in \mathcal{R}$, then $\{1_A \in C\} \in \{\emptyset, A, A^c, \Omega\}$ and $\{1_B \in D\} \in \{\emptyset, B, B^c, \Omega\}$, so there are 16 things to check. When either set involved is $\emptyset$ or Ω , the equality holds, so there are only four cases to worry about and they are all covered by (i). $\square$

In view of the fact that the first definition is a special case of the second, which is a special case of the third, we take things in the opposite order when we say what it means for several things to be independent. We begin by reducing to the case of finitely many objects. An infinite collection of objects (σ -fields, random variables, or sets) is said to be independent if every finite subcollection is.

σ -fields $\mathcal{F}_1, \mathcal{F}_2, \dots, \mathcal{F}_n$ are **independent** if whenever $A_i \in \mathcal{F}_i$ for $i = 1, \dots, n$, we have

$$P\left(\bigcap_{i=1}^n A_i\right) = \prod_{i=1}^n P(A_i)$$

Random variables $X_1, \dots, X_n$ are **independent** if whenever $B_i \in \mathcal{R}$ for $i = 1, \dots, n$ we have

$$P\left(\bigcap_{i=1}^n \{X_i \in B_i\}\right) = \prod_{i=1}^n P(X_i \in B_i)$$

Sets $A_1, \dots, A_n$ are **independent** if whenever $I \subset \{1, \dots, n\}$ we have

$$P\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} P(A_i)$$

At first glance, it might seem that the last definition does not match the other two. However, if you think about it for a minute, you will see that if the indicator variables 1_{A_i} , $1 \leq i \leq n$ are independent and we take $B_i = \{1\}$ for $i \in I$, and $B_i = \mathbf{R}$ for $i \notin I$, then the condition in the definition results. Conversely:

Theorem 2.1.3 *Let $A_1, A_2, \dots, A_n$ be independent. (i) $A_1^c, A_2, \dots, A_n$ are independent; (ii) $1_{A_1}, \dots, 1_{A_n}$ are independent.*

Proof (i) Let $B_1 = A_1^c$ and $B_i = A_i$ for $i > 1$. If $I \subset \{1, \dots, n\}$ does not contain 1, it is clear that $P(\bigcap_{i \in I} B_i) = \prod_{i \in I} P(B_i)$. Suppose now that $1 \in I$ and let $J = I - \{1\}$. Subtracting $P(\bigcap_{i \in I} A_i) = \prod_{i \in I} P(A_i)$ from $P(\bigcap_{i \in J} A_i) = \prod_{i \in J} P(A_i)$ gives $P(A_1^c \cap \bigcap_{i \in J} A_i) = P(A_1^c) \prod_{i \in J} P(A_i)$.

(ii) Iterating (i) we see that if $B_i \in \{A_i, A_i^c\}$, then $B_1, \dots, B_n$ are independent. Thus if $C_i \in \{A_i, A_i^c, \Omega\}$ $P(\bigcap_{i=1}^n C_i) = \prod_{i=1}^n P(C_i)$. The last equality holds trivially if some $C_i = \emptyset$, so noting $1_{A_i} \in \{\emptyset, A_i, A_i^c, \Omega\}$ the desired result follows. $\square$

One of the first things to understand about the definition of independent events is that it is not enough to assume $P(A_i \cap A_j) = P(A_i)P(A_j)$ for all $i \neq j$. A sequence of events $A_1, \dots, A_n$ with the last property is called **pairwise independent**. It is clear that independent events are pairwise independent. The next example shows that the converse is not true.

Example 2.1.4 Let X_1, X_2, X_3 be independent random variables with

$$P(X_i = 0) = P(X_i = 1) = 1/2$$

Let $A_1 = \{X_2 = X_3\}$, $A_2 = \{X_3 = X_1\}$, and $A_3 = \{X_1 = X_2\}$. These events are pairwise independent since if $i \neq j$, then

$$P(A_i \cap A_j) = P(X_1 = X_2 = X_3) = 1/4 = P(A_i)P(A_j)$$

but they are not independent since

$$P(A_1 \cap A_2 \cap A_3) = 1/4 \neq 1/8 = P(A_1)P(A_2)P(A_3)$$

In order to show that random variables X and Y are independent, we have to check that $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$ for all Borel sets A and B . Since there are a lot of Borel sets, our next topic is:

2.1.1 Sufficient Conditions for Independence

Our main result is Theorem 2.1.7. To state that result, we need a definition that generalizes all our earlier definitions.

Collections of sets $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n \subset \mathcal{F}$ are said to be **independent** if whenever $A_i \in \mathcal{A}_i$ and $I \subset \{1, \dots, n\}$, we have $P(\cap_{i \in I} A_i) = \prod_{i \in I} P(A_i)$.

If each collection is a single set, i.e., $\mathcal{A}_i = \{A_i\}$, this definition reduces to the one for sets.

Lemma 2.1.5 *Without loss of generality we can suppose each $\mathcal{A}_i$ contains Ω . In this case, the condition is equivalent to*

$$P\left(\cap_{i=1}^n A_i\right) = \prod_{i=1}^n P(A_i) \quad \text{whenever } A_i \in \mathcal{A}_i$$

since we can set $A_i = \Omega$ for $i \notin I$.

Proof If $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n$ are independent and $\bar{\mathcal{A}}_i = \mathcal{A}_i \cup \{\Omega\}$ then $\bar{\mathcal{A}}_1, \bar{\mathcal{A}}_2, \dots, \bar{\mathcal{A}}_n$ are independent, since if $A_i \in \bar{\mathcal{A}}_i$ and $I = \{j : A_j = \Omega\} \cap I$, $\cap_{i \in I} A_i = \cap_{i \in I} \Omega$. $\square$

The proof of Theorem 2.1.7 is based on Dynkin's $\pi - \lambda$ theorem. To state this result, we need two definitions. We say that $\mathcal{A}$ is a **π -system** if it is closed under intersection, i.e., if $A, B \in \mathcal{A}$, then $A \cap B \in \mathcal{A}$. We say that $\mathcal{L}$ is a **λ -system** if: (i) $\Omega \in \mathcal{L}$. (ii) If $A, B \in \mathcal{L}$ and $A \subset B$, then $B - A \in \mathcal{L}$. (iii) If $A_n \in \mathcal{L}$ and $A_n \uparrow A$, then $A \in \mathcal{L}$.

Theorem 2.1.6 ($\pi - \lambda$ Theorem) *If $\mathcal{P}$ is a π -system and $\mathcal{L}$ is a λ -system that contains $\mathcal{P}$, then $\sigma(\mathcal{P}) \subset \mathcal{L}$.*

The proof is hidden away in Section A.1 of the Appendix.

Theorem 2.1.7 *Suppose $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n$ are independent and each $\mathcal{A}_i$ is a π -system. Then $\sigma(\mathcal{A}_1), \sigma(\mathcal{A}_2), \dots, \sigma(\mathcal{A}_n)$ are independent.*

Proof Let $A_2, \dots, A_n$ be sets with $A_i \in \mathcal{A}_i$, let $F = A_2 \cap \dots \cap A_n$ and let $\mathcal{L} = \{A : P(A \cap F) = P(A)P(F)\}$. Since $P(\Omega \cap F) = P(\Omega)P(F)$, $\Omega \in \mathcal{L}$. To check (ii) of the definition of a λ -system, we note that if $A, B \in \mathcal{L}$ with $A \subset B$ then $(B - A) \cap F = (B \cap F) - (A \cap F)$. So using (i) in Theorem 1.1.1, $A, B \in \mathcal{L}$ and then (i) in Theorem 1.1.1 again:

$$\begin{aligned} P((B - A) \cap F) &= P(B \cap F) - P(A \cap F) = P(B)P(F) - P(A)P(F) \\ &= \{P(B) - P(A)\}P(F) = P(B - A)P(F) \end{aligned}$$

and we have $B - A \in \mathcal{L}$. To check (iii) let $B_k \in \mathcal{L}$ with $B_k \uparrow B$ and note that $(B_k \cap F) \uparrow (B \cap F)$ so using (iii) in Theorem 1.1.1, the fact that $B_k \in \mathcal{L}$, and then (iii) in Theorem 1.1.1 again:

$$P(B \cap F) = \lim_k P(B_k \cap F) = \lim_k P(B_k)P(F) = P(B)P(F)$$

Applying the $\pi - \lambda$ theorem now gives $\mathcal{L} \supset \sigma(\mathcal{A}_1)$. It follows that if $A_1 \in \sigma(\mathcal{A}_1)$ and $A_i \in \mathcal{A}_i$ for $2 \leq i \leq n$, then

$$P(\cap_{i=1}^n A_i) = P(A_1)P(\cap_{i=2}^n A_i) = \prod_{i=1}^n P(A_i)$$

Using Lemma 2.1.5 now, we have:

(*) If $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n$ are independent then $\sigma(\mathcal{A}_1), \mathcal{A}_2, \dots, \mathcal{A}_n$ are independent.

Applying (*) to $\mathcal{A}_2, \dots, \mathcal{A}_n, \sigma(\mathcal{A}_1)$ (which are independent since the definition is unchanged by permuting the order) shows that $\sigma(\mathcal{A}_2), \mathcal{A}_3, \dots, \mathcal{A}_n, \sigma(\mathcal{A}_1)$ are independent, and after n iterations we have the desired result. $\square$

Remark The reader should note that it is not easy to show that if $A, B \in \mathcal{L}$, then $A \cap B \in \mathcal{L}$, or $A \cup B \in \mathcal{L}$, but it is easy to check that if $A, B \in \mathcal{L}$ with $A \subset B$, then $B - A \in \mathcal{L}$.

Having worked to establish Theorem 2.1.7, we get several corollaries.

Theorem 2.1.8 *In order for $X_1, \dots, X_n$ to be independent, it is sufficient that for all $x_1, \dots, x_n \in (-\infty, \infty]$*

$$P(X_1 \leq x_1, \dots, X_n \leq x_n) = \prod_{i=1}^n P(X_i \leq x_i)$$

Proof Let $\mathcal{A}_i$ = the sets of the form $\{X_i \leq x_i\}$. Since

$$\{X_i \leq x\} \cap \{X_i \leq y\} = \{X_i \leq x \wedge y\},$$

where $(x \wedge y)_i = x_i \wedge y_i = \min\{x_i, y_i\}$. $\mathcal{A}_i$ is a π -system. Since we have allowed $x_i = \infty$, $\Omega \in \mathcal{A}_i$. Exercise 1.3.1 implies $\sigma(\mathcal{A}_i) = \sigma(X_i)$, so the result follows from Theorem 2.1.7. $\square$

The last result expresses independence of random variables in terms of their distribution functions. Exercises 2.1.1 and 2.1.2 treat density functions and discrete random variables.

Our next goal is to prove that functions of disjoint collections of independent random variables are independent. See Theorem 2.1.10 for the precise statement. First, we will prove an analogous result for σ -fields.

Theorem 2.1.9 *Suppose $\mathcal{F}_{i,j}, 1 \leq i \leq n, 1 \leq j \leq m(i)$ are independent and let $\mathcal{G}_i = \sigma(\cup_j \mathcal{F}_{i,j})$. Then $\mathcal{G}_1, \dots, \mathcal{G}_n$ are independent.*

Proof Let $\mathcal{A}_i$ be the collection of sets of the form $\cap_j A_{i,j}$ where $A_{i,j} \in \mathcal{F}_{i,j}$. $\mathcal{A}_i$ is a π -system that contains Ω and contains $\cup_j \mathcal{F}_{i,j}$ so Theorem 2.1.7 implies $\sigma(\mathcal{A}_i) = \mathcal{G}_i$ are independent. $\square$

Theorem 2.1.10 *If for $1 \leq i \leq n$, $1 \leq j \leq m(i)$, $X_{i,j}$ are independent and $f_i : \mathbf{R}^{m(i)} \rightarrow \mathbf{R}$ are measurable, then $f_i(X_{i,1}, \dots, X_{i,m(i)})$ are independent.*

Proof Let $\mathcal{F}_{i,j} = \sigma(X_{i,j})$ and $\mathcal{G}_i = \sigma(\cup_j \mathcal{F}_{i,j})$. Since $f_i(X_{i,1}, \dots, X_{i,m(i)}) \in \mathcal{G}_i$, the desired result follows from Theorem 2.1.9 and Exercise 2.1.1. $\square$

A concrete special case of Theorem 2.1.10 that we will use shortly is: if $X_1, \dots, X_n$ are independent, then $X = X_1$ and $Y = X_2 \cdots X_n$ are independent. Later, when we study sums $S_m = X_1 + \cdots + X_m$ of independent random variables $X_1, \dots, X_n$, we will use Theorem 2.1.10 to conclude that if $m < n$, then $S_n - S_m$ is independent of the indicator function of the event $\{\max_{1 \leq k \leq m} S_k > x\}$.

2.1.2 Independence, Distribution, and Expectation

Our next goal is to obtain formulas for the distribution and expectation of independent random variables.

Theorem 2.1.11 *Suppose $X_1, \dots, X_n$ are independent random variables and X_i has distribution μ_i , then $(X_1, \dots, X_n)$ has distribution $\mu_1 \times \cdots \times \mu_n$.*

Proof Using the definitions of (i) $A_1 \times \cdots \times A_n$, (ii) independence, (iii) μ_i , and (iv) $\mu_1 \times \cdots \times \mu_n$

$$\begin{aligned} P((X_1, \dots, X_n) \in A_1 \times \cdots \times A_n) &= P(X_1 \in A_1, \dots, X_n \in A_n) \\ &= \prod_{i=1}^n P(X_i \in A_i) = \prod_{i=1}^n \mu_i(A_i) = \mu_1 \times \cdots \times \mu_n(A_1 \times \cdots \times A_n) \end{aligned}$$

The last formula shows that the distribution of $(X_1, \dots, X_n)$ and the measure $\mu_1 \times \cdots \times \mu_n$ agree on sets of the form $A_1 \times \cdots \times A_n$, a π -system that generates $\mathcal{R}^n$. So Theorem 2.1.6 implies they must agree. $\square$

Theorem 2.1.12 *Suppose X and Y are independent and have distributions μ and ν . If $h : \mathbf{R}^2 \rightarrow \mathbf{R}$ is a measurable function with $h \geq 0$ or $E|h(X, Y)| < \infty$, then*

$$Eh(X, Y) = \iint h(x, y) \mu(dx) \nu(dy)$$

In particular, if $h(x, y) = f(x)g(y)$ where $f, g : \mathbf{R} \rightarrow \mathbf{R}$ are measurable functions with $f, g \geq 0$ or $E|f(X)|$ and $E|g(Y)| < \infty$, then

$$Ef(X)g(Y) = Ef(X) \cdot Eg(Y)$$

Proof Using Theorem 1.6.9 and then Fubini's theorem (Theorem 1.7.2), we have

$$Eh(X, Y) = \int_{\mathbf{R}^2} h d(\mu \times \nu) = \iint h(x, y) \mu(dx) \nu(dy)$$

To prove the second result, we start with the result when $f, g \geq 0$. In this case, using the first result, the fact that $g(y)$ does not depend on x and then Theorem 1.6.9 twice we get

$$\begin{aligned} Ef(X)g(Y) &= \iint f(x)g(y) \mu(dx) \nu(dy) = \int g(y) \int f(x) \mu(dx) \nu(dy) \\ &= \int Ef(X)g(y) \nu(dy) = Ef(X)Eg(Y) \end{aligned}$$

Applying the result for nonnegative f and g to $|f|$ and $|g|$ shows $E|f(X)g(Y)| = E|f(X)|E|g(Y)| < \infty$, and we can repeat the last argument to prove the desired result. $\square$

From Theorem 2.1.12, it is only a small step to:

Theorem 2.1.13 *If $X_1, \dots, X_n$ are independent and have (a) $X_i \geq 0$ for all i , or (b) $E|X_i| < \infty$ for all i , then*

$$E \left(\prod_{i=1}^n X_i \right) = \prod_{i=1}^n EX_i$$

i.e., the expectation on the left exists and has the value given on the right.

Proof $X = X_1$ and $Y = X_2 \cdots X_n$ are independent by Theorem 2.1.10, so taking $f(x) = |x|$ and $g(y) = |y|$, we have $E|X_1 \cdots X_n| = E|X_1|E|X_2 \cdots X_n|$, and it follows by induction that if $1 \leq m \leq n$

$$E|X_m \cdots X_n| = \prod_{i=m}^n E|X_i|$$

If the $X_i \geq 0$, then $|X_i| = X_i$ and the desired result follows from the special case $m = 1$. To prove the result in general, note that the special case $m = 2$ implies $E|Y| = E|X_2 \cdots X_n| < \infty$, so using Theorem 2.1.12 with $f(x) = x$ and $g(y) = y$ shows $E(X_1 \cdots X_n) = EX_1 \cdot E(X_2 \cdots X_n)$, and the desired result follows by induction. $\square$

Example 2.1.14 It can happen that $E(XY) = EX \cdot EY$ without the variables being independent. Suppose the joint distribution of X and Y is given by the following table

		Y		
		1	0	-1
X	1	0	a	0
	0	b	c	b
	-1	0	a	0

where $a, b > 0$, $c \geq 0$, and $2a + 2b + c = 1$. Things are arranged so that $XY \equiv 0$. Symmetry implies $EX = 0$ and $EY = 0$, so $E(XY) = 0 = EXEY$. The random variables are not independent since

$$P(X = 1, Y = 1) = 0 < ab = P(X = 1)P(Y = 1)$$

Two random variables X and Y with $EX^2, EY^2 < \infty$ that have $EXY = EXEY$ are said to be **uncorrelated**. The finite second moments are needed so that we know $E|XY| < \infty$ by the Cauchy-Schwarz inequality.

2.1.3 Sums of Independent Random Variables

Theorem 2.1.15 *If X and Y are independent, $F(x) = P(X \leq x)$, and $G(y) = P(Y \leq y)$, then*

$$P(X + Y \leq z) = \int F(z - y) dG(y)$$

The integral on the right-hand side is called the **convolution** of F and G and is denoted $F * G(z)$. The meaning of $dG(y)$ will be explained in the proof.

Proof Let $h(x, y) = 1_{(x+y \leq z)}$. Let μ and ν be the probability measures with distribution functions F and G . Since for fixed y

$$\int h(x, y) \mu(dx) = \int 1_{(-\infty, z-y]}(x) \mu(dx) = F(z - y)$$

using Theorem 2.1.12 gives

$$\begin{aligned} P(X + Y \leq z) &= \iint 1_{(x+y \leq z)} \mu(dx) \nu(dy) \\ &= \int F(z - y) \nu(dy) = \int F(z - y) dG(y) \end{aligned}$$

The last equality is just a change of notation: We regard $dG(y)$ as a shorthand for “integrate with respect to the measure ν with distribution function G .” $\square$

To treat concrete examples, we need a special case of Theorem 2.1.15.

Theorem 2.1.16 *Suppose that X with density f and Y with distribution function G are independent. Then $X + Y$ has density*

$$h(x) = \int f(x - y) dG(y)$$

When Y has density g , the last formula can be written as

$$h(x) = \int f(x - y) g(y) dy$$

Proof From Theorem 2.1.15, the definition of density function, and Fubini’s theorem (Theorem 1.7.2), which is justified since everything is nonnegative, we get

$$\begin{aligned} P(X + Y \leq z) &= \int F(z - y) dG(y) = \int \int_{-\infty}^z f(x - y) dx dG(y) \\ &= \int_{-\infty}^z \int f(x - y) dG(y) dx \end{aligned}$$

The last equation says that $X + Y$ has density $h(x) = \int f(x - y) dG(y)$. The second formula follows from the first when we recall the meaning of $dG(y)$ given in the proof of Theorem 2.1.15 and use Exercise 1.6.8. $\square$

Theorem 2.1.16 plus some ugly calculus allows us to treat two standard examples. These facts should be familiar from undergraduate probability.

Example 2.1.17 The **gamma density** with parameters α and λ is given by

$$f(x) = \begin{cases} \lambda^\alpha x^{\alpha-1} e^{-\lambda x} / \Gamma(\alpha) & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

where $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$.

Theorem 2.1.18 If $X = \text{gamma}(\alpha, \lambda)$ and $Y = \text{gamma}(\beta, \lambda)$ are independent then $X + Y$ is $\text{gamma}(\alpha + \beta, \lambda)$. Consequently, if $X_1, \dots, X_n$ are independent $\text{exponential}(\lambda)$ r.v.'s, then $X_1 + \dots + X_n$ has a $\text{gamma}(n, \lambda)$ distribution.

Proof Writing $f_{X+Y}(z)$ for the density function of $X + Y$ and using Theorem 2.1.16

$$\begin{aligned} f_{X+Y}(x) &= \int_0^x \frac{\lambda^\alpha (x-y)^{\alpha-1}}{\Gamma(\alpha)} e^{-\lambda(x-y)} \frac{\lambda^\beta y^{\beta-1}}{\Gamma(\beta)} e^{-\lambda y} dy \\ &= \frac{\lambda^{\alpha+\beta} e^{-\lambda x}}{\Gamma(\alpha)\Gamma(\beta)} \int_0^x (x-y)^{\alpha-1} y^{\beta-1} dy \end{aligned}$$

so it suffices to show the integral is $x^{\alpha+\beta-1} \Gamma(\alpha)\Gamma(\beta) / \Gamma(\alpha + \beta)$. To do this, we begin by changing variables $y = xu$, $dy = x du$ to get

$$x^{\alpha+\beta-1} \int_0^1 (1-u)^{\alpha-1} u^{\beta-1} du = \int_0^x (x-y)^{\alpha-1} y^{\beta-1} dy \quad (2.1.1)$$

There are two ways to complete the proof at this point. The soft solution is to note that we have shown that the density $f_{X+Y}(x) = c_{\alpha,\beta} e^{-\lambda x} \lambda^{\alpha+\beta} x^{\alpha+\beta-1}$ where

$$c_{\alpha,\beta} = \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 (1-u)^{\alpha-1} u^{\beta-1} du$$

There is only one norming constant $c_{\alpha,\beta}$ that makes this a probability distribution, so recalling the definition of the beta distribution, we must have $c_{\alpha,\beta} = 1 / \Gamma(\alpha + \beta)$.

The less elegant approach for those of us who cannot remember the definition of the beta is to prove the last equality by calculus. Multiplying each side of the last equation by e^{-x} , integrating from 0 to ∞ , and then using Fubini's theorem on the right we have

$$\begin{aligned} \Gamma(\alpha + \beta) \int_0^1 (1-u)^{\alpha-1} u^{\beta-1} du &= \int_0^\infty \int_0^x y^{\beta-1} e^{-y} (x-y)^{\alpha-1} e^{-(x-y)} dy dx \\ &= \int_0^\infty y^{\beta-1} e^{-y} \int_y^\infty (x-y)^{\alpha-1} e^{-(x-y)} dx dy = \Gamma(\alpha)\Gamma(\beta) \end{aligned}$$

which gives the first result. The second follows from the fact that a $\text{gamma}(1, \lambda)$ is an exponential with parameter λ and induction. $\square$

Example 2.1.19 Normal distribution. In Example 1.6.11, we introduced the normal density with mean μ and variance a ,

$$(2\pi a)^{-1/2} \exp(-(x - \mu)^2 / 2a).$$

Theorem 2.1.20 If $X = \text{normal}(\mu, a)$ and $Y = \text{normal}(v, b)$ are independent, then $X + Y = \text{normal}(\mu + v, a + b)$.

Proof It is enough to prove the result for $\mu = v = 0$. Suppose $Y_1 = \text{normal}(0, a)$ and $Y_2 = \text{normal}(0, b)$. Then Theorem 2.1.16 implies

$$f_{Y_1+Y_2}(z) = \frac{1}{2\pi\sqrt{ab}} \int e^{-x^2/2a} e^{-(z-x)^2/2b} dx$$

Dropping the constant in front, the integral can be rewritten as

$$\begin{aligned} & \int \exp\left(-\frac{bx^2 + ax^2 - 2axz + az^2}{2ab}\right) dx \\ &= \int \exp\left(-\frac{a+b}{2ab} \left\{x^2 - \frac{2a}{a+b}xz + \frac{a}{a+b}z^2\right\}\right) dx \\ &= \int \exp\left(-\frac{a+b}{2ab} \left\{\left(x - \frac{a}{a+b}z\right)^2 + \frac{ab}{(a+b)^2}z^2\right\}\right) dx \end{aligned}$$

since $-\{a/(a+b)\}^2 + \{a/(a+b)\} = ab/(a+b)^2$. Factoring out the term that does not depend on x , the last integral

$$\begin{aligned} &= \exp\left(-\frac{z^2}{2(a+b)}\right) \int \exp\left(-\frac{a+b}{2ab} \left(x - \frac{a}{a+b}z\right)^2\right) dx \\ &= \exp\left(-\frac{z^2}{2(a+b)}\right) \sqrt{2\pi ab/(a+b)} \end{aligned}$$

since the last integral is the normal density with parameters $\mu = az/(a+b)$ and $\sigma^2 = ab/(a+b)$ without its proper normalizing constant. Reintroducing the constant we dropped at the beginning,

$$f_{Y_1+Y_2}(z) = \frac{1}{2\pi\sqrt{ab}} \sqrt{2\pi ab/(a+b)} \exp\left(-\frac{z^2}{2(a+b)}\right) \quad \square$$

2.1.4 Constructing Independent Random Variables

The last question that we have to address before we can study independent random variables is: Do they exist? (If they don't exist, then there is no point in studying them!) If we are given a finite number of distribution functions $F_i, 1 \leq i \leq n$, it is easy to construct independent random variables $X_1, \dots, X_n$ with $P(X_i \leq x) = F_i(x)$. Let $\Omega = \mathbf{R}^n$, $\mathcal{F} = \mathcal{R}^n$, $X_i(\omega_1, \dots, \omega_n) = \omega_i$ (the i th coordinate of $\omega \in \mathbf{R}^n$), and let P be the measure on $\mathcal{R}^n$ that has

$$P((a_1, b_1] \times \dots \times (a_n, b_n]) = (F_1(b_1) - F_1(a_1)) \cdots (F_n(b_n) - F_n(a_n))$$

If μ_i is the measure with distribution function F_i , then $P = \mu_1 \times \dots \times \mu_n$.

To construct an infinite sequence $X_1, X_2, \dots$ of independent random variables with given distribution functions, we want to perform the last construction on the infinite product space

$$\mathbf{R}^{\mathbf{N}} = \{(\omega_1, \omega_2, \dots) : \omega_i \in \mathbf{R}\} = \{\text{functions } \omega : \mathbf{N} \rightarrow \mathbf{R}\}$$

where $\mathbf{N} = \{1, 2, \dots\}$ and $\mathbf{N}$ stands for **natural numbers**. We define $X_i(\omega) = \omega_i$ and we equip $\mathbf{R}^{\mathbf{N}}$ with the product σ -field $\mathcal{R}^{\mathbf{N}}$, which is generated by the **finite dimensional sets** = sets of the form $\{\omega : \omega_i \in B_i, 1 \leq i \leq n\}$, where $B_i \in \mathcal{R}$. It is clear how we want to define P for finite dimensional sets. To assert the existence of a unique extension to $\mathcal{R}^{\mathbf{N}}$, we use Theorem A.3.1:

Theorem 2.1.21 (Kolmogorov's extension theorem) *Suppose we are given probability measures μ_n on $(\mathbf{R}^n, \mathcal{R}^n)$ that are consistent, that is,*

$$\mu_{n+1}((a_1, b_1] \times \cdots \times (a_n, b_n] \times \mathbf{R}) = \mu_n((a_1, b_1] \times \cdots \times (a_n, b_n])$$

Then there is a unique probability measure P on $(\mathbf{R}^{\mathbf{N}}, \mathcal{R}^{\mathbf{N}})$ with

$$P(\omega : \omega_i \in (a_i, b_i], 1 \leq i \leq n) = \mu_n((a_1, b_1] \times \cdots \times (a_n, b_n])$$

In what follows we will need to construct sequences of random variables that take values in other measurable spaces $(S, \mathcal{S})$. Unfortunately, Theorem 2.1.21 is not valid for arbitrary measurable spaces. The first example (on an infinite product of different spaces $\Omega_1 \times \Omega_2 \times \dots$) was constructed by Andersen and Jessen (1948). (See Halmos (1950) p. 214 or Neveu (1965) p. 84.) For an example in which all the spaces Ω_i are the same see Wegner (1973). Fortunately, there is a class of spaces that is adequate for all of our results and for which the generalization of Kolmogorov's theorem is trivial.

$(S, \mathcal{S})$ is said to be **nice** if there is a 1-1 map φ from S into $\mathbf{R}$ so that φ and φ^{-1} are both measurable.

Such spaces are often called **standard Borel spaces**, but we already have too many things named after Borel. The next result shows that most spaces arising in applications are nice.

Theorem 2.1.22 *If S is a Borel subset of a complete separable metric space M , and $\mathcal{S}$ is the collection of Borel subsets of S , then $(S, \mathcal{S})$ is nice.*

Proof We begin with the special case $S = [0, 1)^{\mathbf{N}}$ with metric

$$\rho(x, y) = \sum_{n=1}^{\infty} |x_n - y_n|/2^n$$

If $x = (x^1, x^2, x^3, \dots)$, expand each component in binary $x^j = .x_1^j x_2^j x_3^j \dots$ (taking the expansion with an infinite number of 0's). Let

$$\varphi_o(x) = .x_1^1 x_2^1 x_1^2 x_3^1 x_2^2 x_4^1 x_3^2 x_2^3 x_1^4 \dots$$

To treat the general case, we observe that by letting

$$d(x, y) = \rho(x, y)/(1 + \rho(x, y))$$

(for more details, see Exercise 2.1.3), we can suppose that the metric has $d(x, y) < 1$ for all x, y . Let $q_1, q_2, \dots$ be a countable dense set in S . Let

$$\psi(x) = (d(x, q_1), d(x, q_2), \dots).$$

$\psi : S \rightarrow [0, 1)^{\mathbf{N}}$ is continuous and 1-1. $\varphi_o \circ \psi$ gives the desired mapping. □

Caveat emptor. The proof here is somewhat light when it comes to details. For a more comprehensive discussion, see section 13.1 of Dudley (1989). An interesting consequence of the analysis there is that for Borel subsets of a complete separable metric space the continuum hypothesis is true: i.e., all sets are either finite, countably infinite, or have the cardinality of the real numbers.

Exercises

2.1.1 Suppose $(X_1, \dots, X_n)$ has density $f(x_1, x_2, \dots, x_n)$, that is

$$P((X_1, X_2, \dots, X_n) \in A) = \int_A f(x) dx \text{ for } A \in \mathcal{R}^n$$

If $f(x)$ can be written as $g_1(x_1) \cdots g_n(x_n)$ where the $g_m \geq 0$ are measurable, then $X_1, X_2, \dots, X_n$ are independent. Note that the g_m are not assumed to be probability densities.

2.1.2 Suppose $X_1, \dots, X_n$ are random variables that take values in countable sets $S_1, \dots, S_n$. Then in order for $X_1, \dots, X_n$ to be independent, it is sufficient that whenever $x_i \in S_i$

$$P(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n P(X_i = x_i)$$

2.1.3 Let $\rho(x, y)$ be a metric. (i) Suppose h is differentiable with $h(0) = 0$, $h'(x) > 0$ for $x > 0$, and $h'(x)$ decreasing on $[0, \infty)$. Then $h(\rho(x, y))$ is a metric. (ii) $h(x) = x/(x+1)$ satisfies the hypotheses in (i).

2.1.4 Let $\Omega = (0, 1)$, $\mathcal{F}$ = Borel sets, P = Lebesgue measure. $X_n(\omega) = \sin(2\pi n\omega)$, $n = 1, 2, \dots$ are uncorrelated but not independent.

2.1.5 (i) Show that if X and Y are independent with distributions μ and ν , then

$$P(X + Y = 0) = \sum_y \mu(\{-y\})\nu(\{y\})$$

(ii) Conclude that if X has continuous distribution $P(X = Y) = 0$.

2.1.6 Prove directly from the definition that if X and Y are independent and f and g are measurable functions, then $f(X)$ and $g(Y)$ are independent.

2.1.7 Let $K \geq 3$ be a prime and let X and Y be independent random variables that are uniformly distributed on $\{0, 1, \dots, K-1\}$. For $0 \leq n < K$, let $Z_n = X + nY \bmod K$. Show that $Z_0, Z_1, \dots, Z_{K-1}$ are **pairwise independent**, i.e., each pair is independent. They are not independent because if we know the values of two of the variables then we know the values of all the variables.

2.1.8 Find four random variables taking values in $\{-1, 1\}$ so that any three are independent but all four are not. Hint: Consider products of independent random variables.

2.1.9 Let $\Omega = \{1, 2, 3, 4\}$, $\mathcal{F}$ = all subsets of Ω , and $P(\{i\}) = 1/4$. Give an example of two collections of sets $\mathcal{A}_1$ and $\mathcal{A}_2$ that are independent but whose generated σ -fields are not.

2.1.10 Show that if X and Y are independent, integer-valued random variables, then

$$P(X + Y = n) = \sum_m P(X = m)P(Y = n - m)$$

2.1.11 In Example 1.6.13, we introduced the Poisson distribution with parameter λ , which is given by $P(Z = k) = e^{-\lambda} \lambda^k / k!$ for $k = 0, 1, 2, \dots$. Use the previous exercise to show that if $X = \text{Poisson}(\lambda)$ and $Y = \text{Poisson}(\mu)$ are independent, then $X + Y = \text{Poisson}(\lambda + \mu)$.

2.1.12 X is said to have a Binomial(n, p) distribution if

$$P(X = m) = \binom{n}{m} p^m (1 - p)^{n-m}$$

(i) Show that if $X = \text{Binomial}(n, p)$ and $Y = \text{Binomial}(m, p)$ are independent then $X + Y = \text{Binomial}(n + m, p)$. (ii) Look at Example 1.6.12 and use induction to conclude that the sum of n independent Bernoulli(p) random variables is Binomial(n, p).

2.1.13 It should not be surprising that the distribution of $X + Y$ can be $F * G$ without the random variables being independent. Suppose $X, Y \in \{0, 1, 2\}$ and take each value with probability $1/3$. (a) Find the distribution of $X + Y$ assuming X and Y are independent. (b) Find all the joint distributions (X, Y) so that the distribution of $X + Y$ is the same as the answer to (a).

2.1.14 Let $X, Y \geq 0$ be independent with distribution functions F and G . Find the distribution function of XY .

2.1.15 If we want an infinite sequence of coin tossings, we do not have to use Kolmogorov's theorem. Let Ω be the unit interval $(0, 1)$ equipped with the Borel sets $\mathcal{F}$ and Lebesgue measure P . Let $Y_n(\omega) = 1$ if $[2^n \omega]$ is odd and 0 if $[2^n \omega]$ is even. Show that $Y_1, Y_2, \dots$ are independent with $P(Y_k = 0) = P(Y_k = 1) = 1/2$.

2.2 Weak Laws of Large Numbers

In this section, we will prove several “weak laws of large numbers.” The first order of business is to define the mode of convergence that appears in the conclusions of the theorems. We say that Y_n converges to Y **in probability** if for all $\epsilon > 0$, $P(|Y_n - Y| > \epsilon) \rightarrow 0$ as $n \rightarrow \infty$.

2.2.1 L^2 Weak Laws

Our first set of weak laws come from computing variances and using Chebyshev's inequality. Extending a definition given in Example 2.1.14 for two random variables, a family of random variables $X_i, i \in I$ with $EX_i^2 < \infty$ is said to be **uncorrelated** if we have

$$E(X_i X_j) = EX_i EX_j \quad \text{whenever } i \neq j$$

The key to our weak law for uncorrelated random variables, Theorem 2.2.3, is:

Theorem 2.2.1 Let $X_1, \dots, X_n$ have $E(X_i^2) < \infty$ and be uncorrelated. Then

$$\text{var}(X_1 + \dots + X_n) = \text{var}(X_1) + \dots + \text{var}(X_n)$$

where $\text{var}(Y) = \text{the variance of } Y$.

Proof Let $\mu_i = EX_i$ and $S_n = \sum_{i=1}^n X_i$. Since $ES_n = \sum_{i=1}^n \mu_i$, using the definition of the variance, writing the square of the sum as the product of two copies of the sum, and then expanding, we have

$$\begin{aligned} \text{var}(S_n) &= E(S_n - ES_n)^2 = E\left(\sum_{i=1}^n (X_i - \mu_i)\right)^2 \\ &= E\left(\sum_{i=1}^n \sum_{j=1}^n (X_i - \mu_i)(X_j - \mu_j)\right) \\ &= \sum_{i=1}^n E(X_i - \mu_i)^2 + 2 \sum_{i=1}^n \sum_{j=1}^{i-1} E((X_i - \mu_i)(X_j - \mu_j)) \end{aligned}$$

where in the last equality we have separated out the diagonal terms $i = j$ and used the fact that the sum over $1 \leq i < j \leq n$ is the same as the sum over $1 \leq j < i \leq n$.

The first sum is $\text{var}(X_1) + \dots + \text{var}(X_n)$ so we want to show that the second sum is zero. To do this, we observe

$$\begin{aligned} E((X_i - \mu_i)(X_j - \mu_j)) &= EX_i X_j - \mu_i EX_j - \mu_j EX_i + \mu_i \mu_j \\ &= EX_i X_j - \mu_i \mu_j = 0 \end{aligned}$$

since X_i and X_j are uncorrelated. □

In words, Theorem 2.2.1 says that for uncorrelated random variables the variance of the sum is the sum of the variances. The second ingredient in our proof of Theorem 2.2.3 is the following consequence of (1.6.4):

$$\text{var}(cY) = c^2 \text{var}(Y)$$

The third and final ingredient is

Lemma 2.2.2 If $p > 0$ and $E|Z_n|^p \rightarrow 0$, then $Z_n \rightarrow 0$ in probability.

Proof Chebyshev's inequality, Theorem 1.6.4, with $\varphi(x) = x^p$ and $X = |Z_n|$ implies that if $\epsilon > 0$ then $P(|Z_n| \geq \epsilon) \leq \epsilon^{-p} E|Z_n|^p \rightarrow 0$. □

We can now easily prove:

Theorem 2.2.3 (L^2 weak law) Let $X_1, X_2, \dots$ be uncorrelated random variables with $EX_i = \mu$ and $\text{var}(X_i) \leq C < \infty$. If $S_n = X_1 + \dots + X_n$, then as $n \rightarrow \infty$, $S_n/n \rightarrow \mu$ in L^2 and in probability.

Proof To prove L^2 convergence, observe that $E(S_n/n) = \mu$, so

$$E(S_n/n - \mu)^2 = \text{var}(S_n/n) = \frac{1}{n^2} (\text{var}(X_1) + \dots + \text{var}(X_n)) \leq \frac{Cn}{n^2} \rightarrow 0$$

To conclude there is also convergence in probability, we apply the Lemma 2.2.2 to $Z_n = S_n/n - \mu$. □

The most important special case of Theorem 2.2.3 occurs when $X_1, X_2, \dots$ are independent random variables that all have the same distribution. In the jargon, they are **independent and identically distributed** or **i.i.d.** for short. Theorem 2.2.3 tells us in this case that if $EX_i^2 < \infty$ then S_n/n converges to $\mu = EX_i$ in probability as $n \rightarrow \infty$. In Theorem 2.2.14, we will see that $E|X_i| < \infty$ is sufficient for the last conclusion, but for the moment we will concern ourselves with consequences of the weaker result.

Our first application is to a situation that on the surface has nothing to do with randomness.

Example 2.2.4 (Polynomial approximation) Let f be a continuous function on $[0,1]$, and let

$$f_n(x) = \sum_{m=0}^n \binom{n}{m} x^m (1-x)^{n-m} f(m/n) \quad \text{where} \quad \binom{n}{m} = \frac{n!}{m!(n-m)!}$$

be the **Bernstein polynomial of degree n** associated with f . Then as $n \rightarrow \infty$

$$\sup_{x \in [0,1]} |f_n(x) - f(x)| \rightarrow 0$$

Proof First observe that if S_n is the sum of n independent random variables with $P(X_i = 1) = p$ and $P(X_i = 0) = 1 - p$, then $EX_i = p$, $\text{var}(X_i) = p(1-p)$ and

$$P(S_n = m) = \binom{n}{m} p^m (1-p)^{n-m}$$

so $Ef(S_n/n) = f_n(p)$. Theorem 2.2.3 tells us that as $n \rightarrow \infty$, $S_n/n \rightarrow p$ in probability. The last two observations motivate the definition of $f_n(p)$, but to prove the desired conclusion we have to use the proof of Theorem 2.2.3 rather than the result itself.

Combining the proof of Theorem 2.2.3 with our formula for the variance of X_i and the fact that $p(1-p) \leq 1/4$ when $p \in [0,1]$, we have

$$P(|S_n/n - p| > \delta) \leq \frac{\text{var}(S_n/n)}{\delta^2} = \frac{p(1-p)}{n\delta^2} \leq \frac{1}{4n\delta^2}$$

To conclude now that $Ef(S_n/n) \rightarrow f(p)$, let $M = \sup_{x \in [0,1]} |f(x)|$, let $\epsilon > 0$, and pick $\delta > 0$ so that if $|x - y| < \delta$, then $|f(x) - f(y)| < \epsilon$. (This is possible since a continuous function is uniformly continuous on each bounded interval.) Now, using Jensen's inequality, Theorem 1.6.2, gives

$$|Ef(S_n/n) - f(p)| \leq E|f(S_n/n) - f(p)| \leq \epsilon + 2MP(|S_n/n - p| > \delta)$$

Letting $n \rightarrow \infty$, we have $\limsup_{n \rightarrow \infty} |Ef(S_n/n) - f(p)| \leq \epsilon$, but ϵ is arbitrary, so this gives the desired result. $\square$

Our next result is for comic relief.

Example 2.2.5 (A high-dimensional cube is almost the boundary of a ball) Let $X_1, X_2, \dots$ be independent and uniformly distributed on $(-1,1)$. Let $Y_i = X_i^2$, which are independent since they are functions of independent random variables. $EY_i = 1/3$ and $\text{var}(Y_i) \leq EY_i^2 \leq 1$, so Theorem 2.2.3 implies

$$(X_1^2 + \dots + X_n^2)/n \rightarrow 1/3 \quad \text{in probability as } n \rightarrow \infty$$

Let $A_{n,\epsilon} = \{x \in \mathbf{R}^n : (1-\epsilon)\sqrt{n/3} < |x| < (1+\epsilon)\sqrt{n/3}\}$, where $|x| = (x_1^2 + \cdots + x_n^2)^{1/2}$. If we let $|S|$ denote the Lebesgue measure of S , then the last conclusion implies that for any $\epsilon > 0$, $|A_{n,\epsilon} \cap (-1,1)^n|/2^n \rightarrow 1$, or, in words, most of the volume of the cube $(-1,1)^n$ comes from $A_{n,\epsilon}$, which is almost the boundary of the ball of radius $\sqrt{n/3}$.

2.2.2 Triangular Arrays

Many classical limit theorems in probability concern arrays $X_{n,k}$, $1 \leq k \leq n$ of random variables and investigate the limiting behavior of their row sums $S_n = X_{n,1} + \cdots + X_{n,n}$. In most cases, we assume that the random variables on each row are independent, but for the next trivial (but useful) result we do not need that assumption. Indeed, here S_n can be any sequence of random variables.

Theorem 2.2.6 *Let $\mu_n = ES_n$, $\sigma_n^2 = \text{var}(S_n)$. If $\sigma_n^2/b_n^2 \rightarrow 0$ then*

$$\frac{S_n - \mu_n}{b_n} \rightarrow 0 \quad \text{in probability}$$

Proof Our assumptions imply $E((S_n - \mu_n)/b_n)^2 = b_n^{-2} \text{var}(S_n) \rightarrow 0$, so the desired conclusion follows from Lemma 2.2.2. $\square$

We will now give three applications of Theorem 2.2.6. For these three examples, the following calculation is useful:

$$\begin{aligned} \sum_{m=1}^n \frac{1}{m} &\geq \int_1^n \frac{dx}{x} \geq \sum_{m=2}^n \frac{1}{m} \\ \log n &\leq \sum_{m=1}^n \frac{1}{m} \leq 1 + \log n \end{aligned} \tag{2.2.1}$$

Example 2.2.7 (Coupon collector's problem) Let $X_1, X_2, \dots$ be i.i.d. uniform on $\{1, 2, \dots, n\}$. To motivate the name, think of collecting baseball cards (or coupons). Suppose that the i th item we collect is chosen at random from the set of possibilities and is independent of the previous choices. Let $\tau_k^n = \inf\{m : |\{X_1, \dots, X_m\}| = k\}$ be the first time we have k different items. In this problem, we are interested in the asymptotic behavior of $T_n = \tau_n^n$, the time to collect a complete set. It is easy to see that $\tau_1^n = 1$. To make later formulas work out nicely, we will set $\tau_0^n = 0$. For $1 \leq k \leq n$, $X_{n,k} \equiv \tau_k^n - \tau_{k-1}^n$ represents the time to get a choice different from our first $k-1$, so $X_{n,k}$ has a geometric distribution with parameter $1 - (k-1)/n$ and is independent of the earlier waiting times $X_{n,j}$, $1 \leq j < k$. Example 1.6.14 tells us that if X has a geometric distribution with parameter p then $EX = 1/p$ and $\text{var}(X) \leq 1/p^2$. Using the linearity of expected value, bounds on $\sum_{m=1}^n 1/m$ in (2.2.1), and Theorem 2.2.1 we see that

$$\begin{aligned} ET_n &= \sum_{k=1}^n \left(1 - \frac{k-1}{n}\right)^{-1} = n \sum_{m=1}^n m^{-1} \sim n \log n \\ \text{var}(T_n) &\leq \sum_{k=1}^n \left(1 - \frac{k-1}{n}\right)^{-2} = n^2 \sum_{m=1}^n m^{-2} \leq n^2 \sum_{m=1}^{\infty} m^{-2} \end{aligned}$$

Taking $b_n = n \log n$ and using Theorem 2.2.6, it follows that

$$\frac{T_n - n \sum_{m=1}^n m^{-1}}{n \log n} \rightarrow 0 \quad \text{in probability}$$

and hence $T_n/(n \log n) \rightarrow 1$ in probability.

For a concrete example, take $n = 365$, i.e., we are interested in the number of people we need to meet until we have seen someone with every birthday. In this case, the limit theorem says it will take about $365 \log 365 = 2,153.46$ tries to get a complete set. Note that the number of trials is 5.89 times the number of birthdays.

Example 2.2.8 (Random permutations) Let Ω_n consist of the $n!$ permutations (i.e., one-to-one mappings from $\{1, \dots, n\}$ onto $\{1, \dots, n\}$) and make this into a probability space by assuming all the permutations are equally likely. This application of the weak law concerns the cycle structure of a random permutation π , so we begin by describing the decomposition of a permutation into cycles. Consider the sequence $1, \pi(1), \pi(\pi(1)), \dots$. Eventually, $\pi^k(1) = 1$. When it does, we say the first cycle is completed and has length k . To start the second cycle, we pick the smallest integer i not in the first cycle and look at $i, \pi(i), \pi(\pi(i)), \dots$ until we come back to i . We repeat the construction until all the elements are accounted for. For example, if the permutation is

$$\begin{array}{cccccccc} i & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ \pi(i) & 3 & 9 & 6 & 8 & 2 & 1 & 5 & 4 & 7 \end{array}$$

then the cycle decomposition is $(136)(2975)(48)$.

Let $X_{n,k} = 1$ if a right parenthesis occurs after the k th number in the decomposition, $X_{n,k} = 0$ otherwise, and let $S_n = X_{n,1} + \dots + X_{n,n}$ = the number of cycles. (In the example, $X_{9,3} = X_{9,7} = X_{9,9} = 1$, and the other $X_{9,m} = 0$.) I claim that

Lemma 2.2.9 $X_{n,1}, \dots, X_{n,n}$ are independent and $P(X_{n,j} = 1) = \frac{1}{n-j+1}$.

Intuitively, this is true since, independent of what has happened so far, there are $n - j + 1$ values that have not appeared in the range, and only one of them will complete the cycle.

Proof To prove this, it is useful to generate the permutation in a special way. Let $i_1 = 1$. Pick j_1 at random from $\{1, \dots, n\}$ and let $\pi(i_1) = j_1$. If $j_1 \neq 1$, let $i_2 = j_1$. If $j_1 = 1$, let $i_2 = 2$. In either case, pick j_2 at random from $\{1, \dots, n\} - \{j_1\}$. In general, if $i_1, j_1, \dots, i_{k-1}, j_{k-1}$ have been selected and we have set $\pi(i_\ell) = j_\ell$ for $1 \leq \ell < k$, then (a) if $j_{k-1} \in \{i_1, \dots, i_{k-1}\}$ so a cycle has just been completed, we let $i_k = \inf(\{1, \dots, n\} - \{i_1, \dots, i_{k-1}\})$ and (b) if $j_{k-1} \notin \{i_1, \dots, i_{k-1}\}$, we let $i_k = j_{k-1}$. In either case, we pick j_k at random from $\{1, \dots, n\} - \{j_1, \dots, j_{k-1}\}$ and let $\pi(i_k) = j_k$.

The construction here is tedious to write out, or to read, but now I can claim with a clear conscience that $X_{n,1}, \dots, X_{n,n}$ are independent and $P(X_{n,k} = 1) = 1/(n - k + 1)$ since when we pick j_k there are $n - k + 1$ values in $\{1, \dots, n\} - \{j_1, \dots, j_{k-1}\}$ and only one of them will complete the cycle. $\square$

To check the conditions of Theorem 2.2.6, now note

$$\begin{aligned} ES_n &= 1/n + 1/(n-1) + \dots + 1/2 + 1 \\ \text{var}(S_n) &= \sum_{k=1}^n \text{var}(X_{n,k}) \leq \sum_{k=1}^n E(X_{n,k}^2) = \sum_{k=1}^n E(X_{n,k}) = ES_n \end{aligned}$$

where the results on the second line follow from Theorem 2.2.1, the fact that $\text{var}(Y) \leq EY^2$, and $X_{n,k}^2 = X_{n,k}$. Now $ES_n \sim \log n$, so if $b_n = (\log n)^{.5+\epsilon}$ with $\epsilon > 0$, the conditions of Theorem 2.2.6 are satisfied and it follows that

$$\frac{S_n - \sum_{m=1}^n m^{-1}}{(\log n)^{.5+\epsilon}} \rightarrow 0 \quad \text{in probability} \quad (*)$$

Taking $\epsilon = 0.5$, we have that $S_n / \log n \rightarrow 1$ in probability, but $(*)$ says more. We will see in Example 3.4.11 that $(*)$ is false if $\epsilon = 0$.

Example 2.2.10 (An occupancy problem) Suppose we put r balls at random in n boxes, i.e., all n^r assignments of balls to boxes have equal probability. Let A_i be the event that the i th box is empty and N_n = the number of empty boxes. It is easy to see that

$$P(A_i) = (1 - 1/n)^r \quad \text{and} \quad EN_n = n(1 - 1/n)^r$$

A little calculus (take logarithms) shows that if $r/n \rightarrow c$, $EN_n/n \rightarrow e^{-c}$. (For a proof, see Lemma 3.1.1.) To compute the variance of N_n , we observe that

$$\begin{aligned} EN_n^2 &= E \left(\sum_{m=1}^n 1_{A_m} \right)^2 = \sum_{1 \leq k, m \leq n} P(A_k \cap A_m) \\ \text{var}(N_n) &= EN_n^2 - (EN_n)^2 = \sum_{1 \leq k, m \leq n} P(A_k \cap A_m) - P(A_k)P(A_m) \\ &= n(n-1)\{(1 - 2/n)^r - (1 - 1/n)^{2r}\} + n\{(1 - 1/n)^r - (1 - 1/n)^{2r}\} \end{aligned}$$

The first term comes from $k \neq m$ and the second from $k = m$. Since $(1 - 2/n)^r \rightarrow e^{-2c}$ and $(1 - 1/n)^r \rightarrow e^{-c}$, it follows easily from the last formula that $\text{var}(N_n/n) = \text{var}(N_n)/n^2 \rightarrow 0$. Taking $b_n = n$ in Theorem 2.2.6, now we have

$$N_n/n \rightarrow e^{-c} \quad \text{in probability}$$

2.2.3 Truncation

To truncate a random variable X at level M means considering

$$\bar{X} = X 1_{(|X| \leq M)} = \begin{cases} X & \text{if } |X| \leq M \\ 0 & \text{if } |X| > M \end{cases}$$

To extend the weak law to random variables without a finite second moment, we will truncate and then use Chebyshev's inequality. We begin with a very general but also very useful result. Its proof is easy because we have assumed what we need for the proof. Later we will have to work a little to verify the assumptions in special cases, but the general result serves to identify the essential ingredients in the proof.

Theorem 2.2.11 (Weak law for triangular arrays) For each n let $X_{n,k}$, $1 \leq k \leq n$, be independent. Let $b_n > 0$ with $b_n \rightarrow \infty$, and let $\bar{X}_{n,k} = X_{n,k} 1_{(|X_{n,k}| \leq b_n)}$. Suppose that as $n \rightarrow \infty$

(i) $\sum_{k=1}^n P(|X_{n,k}| > b_n) \rightarrow 0$, and

(ii) $b_n^{-2} \sum_{k=1}^n E \bar{X}_{n,k}^2 \rightarrow 0$.

If we let $S_n = X_{n,1} + \cdots + X_{n,n}$ and put $a_n = \sum_{k=1}^n E \bar{X}_{n,k}$, then

$$(S_n - a_n)/b_n \rightarrow 0 \text{ in probability}$$

Proof Let $\bar{S}_n = \bar{X}_{n,1} + \cdots + \bar{X}_{n,n}$. Clearly,

$$P\left(\left|\frac{S_n - a_n}{b_n}\right| > \epsilon\right) \leq P(S_n \neq \bar{S}_n) + P\left(\left|\frac{\bar{S}_n - a_n}{b_n}\right| > \epsilon\right)$$

To estimate the first term, we note that

$$P(S_n \neq \bar{S}_n) \leq P\left(\bigcup_{k=1}^n \{\bar{X}_{n,k} \neq X_{n,k}\}\right) \leq \sum_{k=1}^n P(|X_{n,k}| > b_n) \rightarrow 0$$

by (i). For the second term, we note that Chebyshev's inequality, $a_n = E \bar{S}_n$, Theorem 2.2.1, and $\text{var}(X) \leq EX^2$ imply

$$\begin{aligned} P\left(\left|\frac{\bar{S}_n - a_n}{b_n}\right| > \epsilon\right) &\leq \epsilon^{-2} E \left|\frac{\bar{S}_n - a_n}{b_n}\right|^2 = \epsilon^{-2} b_n^{-2} \text{var}(\bar{S}_n) \\ &= (b_n \epsilon)^{-2} \sum_{k=1}^n \text{var}(\bar{X}_{n,k}) \leq (b_n \epsilon)^{-2} \sum_{k=1}^n E(\bar{X}_{n,k})^2 \rightarrow 0 \end{aligned}$$

by (ii), and the proof is complete. $\square$

From Theorem 2.2.11, we get the following result for a single sequence.

Theorem 2.2.12 (Weak law of large numbers) *Let $X_1, X_2, \dots$ be i.i.d. with*

$$x P(|X_i| > x) \rightarrow 0 \quad \text{as } x \rightarrow \infty$$

Let $S_n = X_1 + \cdots + X_n$ and let $\mu_n = E(X_1 1_{(|X_1| \leq n)})$. Then $S_n/n - \mu_n \rightarrow 0$ in probability.

Remark The assumption in the theorem is necessary for the existence of constants a_n so that $S_n/n - a_n \rightarrow 0$. See Feller (1971) pp. 234–236 for a proof.

Proof We will apply Theorem 2.2.11 with $X_{n,k} = X_k$ and $b_n = n$. To check (i), we note

$$\sum_{k=1}^n P(|X_{n,k}| > n) = n P(|X_i| > n) \rightarrow 0$$

by assumption. To check (ii), we need to show $n^{-2} \cdot n E \bar{X}_{n,1}^2 \rightarrow 0$. To do this, we need the following result, which will be useful several times later.

Lemma 2.2.13 *If $Y \geq 0$ and $p > 0$ then $E(Y^p) = \int_0^\infty p y^{p-1} P(Y > y) dy$.*

Proof Using the definition of expected value, Fubini's theorem (for nonnegative random variables), and then calculating the resulting integrals gives

$$\begin{aligned}
\int_0^\infty py^{p-1} P(Y > y) dy &= \int_0^\infty \int_\Omega py^{p-1} 1_{(Y>y)} dP dy \\
&= \int_\Omega \int_0^\infty py^{p-1} 1_{(Y>y)} dy dP \\
&= \int_\Omega \int_0^Y py^{p-1} dy dP = \int_\Omega Y^p dP = EY^p
\end{aligned}$$

which is the desired result. $\square$

Returning to the proof of Theorem 2.2.12, we observe that Lemma 2.2.13 and the fact that $\bar{X}_{n,1} = X_1 1_{(|X_1| \leq n)}$ imply

$$E(\bar{X}_{n,1}^2) = \int_0^\infty 2yP(|\bar{X}_{n,1}| > y) dy \leq \int_0^n 2yP(|X_1| > y) dy$$

since $P(|\bar{X}_{n,1}| > y) = 0$ for $y \geq n$ and $= P(|X_1| > y) - P(|X_1| > n)$ for $y \leq n$. We claim that $yP(|X_1| > y) \rightarrow 0$ implies

$$E(\bar{X}_{n,1}^2)/n = \frac{1}{n} \int_0^n 2yP(|X_1| > y) dy \rightarrow 0$$

as $n \rightarrow \infty$. Intuitively, this holds since the right-hand side is the average of $g(y) = 2yP(|X_1| > y)$ over $[0, n]$ and $g(y) \rightarrow 0$ as $y \rightarrow \infty$. To spell out the details, note that $0 \leq g(y) \leq 2y$ and $g(y) \rightarrow 0$ as $y \rightarrow \infty$, so we must have $M = \sup g(y) < \infty$. Let $g_n(y) = g(ny)$. Since g_n is bounded and $g_n(y) \rightarrow 0$ a.s.,

$$(1/n) \int_0^n g(y) dy = \int_0^1 g_n(x) dx \rightarrow 0$$

which completes the proof. $\square$

Remark Applying Lemma 2.2.13 with $p = 1 - \epsilon$ and $\epsilon > 0$, we see that $xP(|X_1| > x) \rightarrow 0$ implies $E|X_1|^{1-\epsilon} < \infty$, so the assumption in Theorem 2.2.12 is not much weaker than finite mean.

Finally, we have the weak law in its most familiar form.

Theorem 2.2.14 *Let $X_1, X_2, \dots$ be i.i.d. with $E|X_i| < \infty$. Let $S_n = X_1 + \dots + X_n$ and let $\mu = EX_1$. Then $S_n/n \rightarrow \mu$ in probability.*

Proof Two applications of the dominated convergence theorem imply

$$\begin{aligned}
xP(|X_1| > x) &\leq E(|X_1| 1_{(|X_1| > x)}) \rightarrow 0 \quad \text{as } x \rightarrow \infty \\
\mu_n &= E(X_1 1_{(|X_1| \leq n)}) \rightarrow E(X_1) = \mu \quad \text{as } n \rightarrow \infty
\end{aligned}$$

Using Theorem 2.2.12, we see that if $\epsilon > 0$ then $P(|S_n/n - \mu_n| > \epsilon/2) \rightarrow 0$. Since $\mu_n \rightarrow \mu$, it follows that $P(|S_n/n - \mu| > \epsilon) \rightarrow 0$. $\square$

Example 2.2.15 For an example where the weak law does not hold, suppose $X_1, X_2, \dots$ are independent and have a **Cauchy distribution**:

$$P(X_i \leq x) = \int_{-\infty}^x \frac{dt}{\pi(1+t^2)}$$

As $x \rightarrow \infty$,

$$P(|X_1| > x) = 2 \int_x^\infty \frac{dt}{\pi(1+t^2)} \sim \frac{2}{\pi} \int_x^\infty t^{-2} dt = \frac{2}{\pi} x^{-1}$$

From the necessity of this condition, we can conclude that there is no sequence of constants μ_n so that $S_n/n - \mu_n \rightarrow 0$. We will see later that S_n/n always has the same distribution as X_1 (See Exercise 3.3.6.).

As the next example shows, we can have a weak law in some situations in which $E|X| = \infty$.

Example 2.2.16 (The “St. Petersburg paradox.”) Let $X_1, X_2, \dots$ be independent random variables with

$$P(X_i = 2^j) = 2^{-j} \quad \text{for } j \geq 1$$

In words, you win 2^j dollars if it takes j tosses to get a heads. The paradox here is that $EX_1 = \infty$, but you clearly wouldn't pay an infinite amount to play this game. An application of Theorem 2.2.11 will tell us how much we should pay to play the game n times.

In this example, $X_{n,k} = X_k$. To apply Theorem 2.2.11, we have to pick b_n . To do this, we are guided by the principle that in checking (ii) we want to take b_n as small as we can and have (i) hold. With this in mind, we observe that if m is an integer

$$P(X_1 \geq 2^m) = \sum_{j=m}^{\infty} 2^{-j} = 2^{-m+1}$$

Let $m(n) = \log_2 n + K(n)$ where $K(n) \rightarrow \infty$ and is chosen so that $m(n)$ is an integer (and hence the displayed formula is valid). Letting $b_n = 2^{m(n)}$, we have

$$nP(X_1 \geq b_n) = n2^{-m(n)+1} = 2^{-K(n)+1} \rightarrow 0$$

proving (i). To check (ii), we observe that if $\bar{X}_{n,k} = X_k 1_{(|X_k| \leq b_n)}$, then

$$E\bar{X}_{n,k}^2 = \sum_{j=1}^{m(n)} 2^{2j} \cdot 2^{-j} \leq 2^{m(n)} \sum_{k=0}^{\infty} 2^{-k} = 2b_n$$

So the expression in (ii) is smaller than $2n/b_n$, which $\rightarrow 0$ since

$$b_n = 2^{m(n)} = n2^{K(n)} \quad \text{and} \quad K(n) \rightarrow \infty$$

The last two steps are to evaluate a_n and to apply Theorem 2.2.11.

$$E\bar{X}_{n,k} = \sum_{j=1}^{m(n)} 2^j 2^{-j} = m(n)$$

so $a_n = nm(n)$. We have $m(n) = \log n + K(n)$ (here and until the end of the example all logs are base 2), so if we pick $K(n)/\log n \rightarrow 0$ then $a_n/n \log n \rightarrow 1$ as $n \rightarrow \infty$. Using Theorem 2.2.11 now, we have

$$\frac{S_n - a_n}{n2^{K(n)}} \rightarrow 0 \quad \text{in probability}$$

If we suppose that $K(n) \leq \log \log n$ for large n , then the last conclusion holds with the denominator replaced by $n \log n$, and it follows that $S_n/(n \log n) \rightarrow 1$ in probability.

Returning to our original question, we see that a fair price for playing n times is $\$ \log_2 n$ per play. When $n = 1024$, this is $\$10$ per play. Nicolas Bernoulli wrote in 1713, "There ought not to exist any even halfway sensible person who would not sell the right of playing the game for 40 ducats (per play)." If the wager were 1 ducat, one would need $2^{40} \approx 10^{12}$ plays to start to break even.

Exercises

2.2.1 Let $X_1, X_2, \dots$ be uncorrelated with $EX_i = \mu_i$ and $\text{var}(X_i)/i \rightarrow 0$ as $i \rightarrow \infty$. Let $S_n = X_1 + \dots + X_n$ and $v_n = ES_n/n$ then as $n \rightarrow \infty$, $S_n/n - v_n \rightarrow 0$ in L^2 and in probability.

2.2.2 The L^2 weak law generalizes immediately to certain dependent sequences. Suppose $EX_n = 0$ and $EX_n X_m \leq r(n - m)$ for $m \leq n$ (no absolute value on the left-hand side!) with $r(k) \rightarrow 0$ as $k \rightarrow \infty$. Show that $(X_1 + \dots + X_n)/n \rightarrow 0$ in probability.

2.2.3 **Monte Carlo integration.** (i) Let f be a measurable function on $[0, 1]$ with $\int_0^1 |f(x)| dx < \infty$. Let $U_1, U_2, \dots$ be independent and uniformly distributed on $[0, 1]$, and let

$$I_n = n^{-1}(f(U_1) + \dots + f(U_n))$$

Show that $I_n \rightarrow I \equiv \int_0^1 f(x) dx$ in probability. (ii) Suppose $\int_0^1 |f(x)|^2 dx < \infty$. Use Chebyshev's inequality to estimate $P(|I_n - I| > a/n^{1/2})$.

2.2.4 Let $X_1, X_2, \dots$ be i.i.d. with $P(X_i = (-1)^k k) = C/k^2 \log k$ for $k \geq 2$ where C is chosen to make the sum of the probabilities = 1. Show that $E|X_i| = \infty$, but there is a finite constant μ so that $S_n/n \rightarrow \mu$ in probability.

2.2.5 Let $X_1, X_2, \dots$ be i.i.d. with $P(X_i > x) = e/x \log x$ for $x \geq e$. Show that $E|X_i| = \infty$, but there is a sequence of constants $\mu_n \rightarrow \infty$ so that $S_n/n - \mu_n \rightarrow 0$ in probability.

2.2.6 (i) Show that if $X \geq 0$ is integer valued $EX = \sum_{n \geq 1} P(X \geq n)$. (ii) Find a similar expression for EX^2 .

2.2.7 Generalize Lemma 2.2.13 to conclude that if $H(x) = \int_{(-\infty, x]} h(y) dy$ with $h(y) \geq 0$, then

$$E H(X) = \int_{-\infty}^{\infty} h(y) P(X \geq y) dy$$

An important special case is $H(x) = \exp(\theta x)$ with $\theta > 0$.

2.2.8 **An unfair "fair game."** Let $p_k = 1/2^k k(k+1)$, $k = 1, 2, \dots$ and $p_0 = 1 - \sum_{k \geq 1} p_k$.

$$\sum_{k=1}^{\infty} 2^k p_k = (1 - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{3}) + \dots = 1$$

so if we let $X_1, X_2, \dots$ be i.i.d. with $P(X_n = -1) = p_0$ and

$$P(X_n = 2^k - 1) = p_k \quad \text{for } k \geq 1$$

then $EX_n = 0$. Let $S_n = X_1 + \dots + X_n$. Use Theorem 2.2.11 with $b_n = 2^{m(n)}$, where $m(n) = \min\{m : 2^{-m}m^{-3/2} \leq n^{-1}\}$ to conclude that

$$S_n/(n/\log_2 n) \rightarrow -1 \text{ in probability}$$

2.2.9 Weak law for positive variables. Suppose $X_1, X_2, \dots$ are i.i.d., $P(0 \leq X_i < \infty) = 1$ and $P(X_i > x) > 0$ for all x . Let $\mu(s) = \int_0^s x dF(x)$ and $\nu(s) = \mu(s)/s(1 - F(s))$. It is known that there exist constants a_n so that $S_n/a_n \rightarrow 1$ in probability, if and only if $\nu(s) \rightarrow \infty$ as $s \rightarrow \infty$. Pick $b_n \geq 1$ so that $n\mu(b_n) = b_n$ (this works for large n), and use Theorem 2.2.11 to prove that the condition is sufficient.

2.3 Borel-Cantelli Lemmas

If A_n is a sequence of subsets of Ω , we let

$$\limsup A_n = \lim_{m \rightarrow \infty} \bigcup_{n=m}^{\infty} A_n = \{\omega \text{ that are in infinitely many } A_n\}$$

(the limit exists since the sequence is decreasing in m) and let

$$\liminf A_n = \lim_{m \rightarrow \infty} \bigcap_{n=m}^{\infty} A_n = \{\omega \text{ that are in all but finitely many } A_n\}$$

(the limit exists since the sequence is increasing in m). The names $\limsup$ and $\liminf$ can be explained by noting that

$$\limsup_{n \rightarrow \infty} 1_{A_n} = 1_{(\limsup A_n)} \quad \liminf_{n \rightarrow \infty} 1_{A_n} = 1_{(\liminf A_n)}$$

It is common to write $\limsup A_n = \{\omega : \omega \in A_n \text{ i.o.}\}$, where i.o. stands for infinitely often. An example that illustrates the use of this notation is: “ $X_n \rightarrow 0$ a.s. if and only if for all $\epsilon > 0$, $P(|X_n| > \epsilon \text{ i.o.}) = 0$.” The reader will see many other examples later.

Exercise 2.3.1 Prove that $P(\limsup A_n) \geq \limsup P(A_n)$ and $P(\liminf A_n) \leq \liminf P(A_n)$.

The next result should be familiar from measure theory, even though its name may not be.

Theorem 2.3.1 (Borel-Cantelli lemma) If $\sum_{n=1}^{\infty} P(A_n) < \infty$, then

$$P(A_n \text{ i.o.}) = 0.$$

Proof Let $N = \sum_k 1_{A_k}$ be the number of events that occur. Fubini’s theorem implies $EN = \sum_k P(A_k) < \infty$, so we must have $N < \infty$ a.s. □

The next result is a typical application of the Borel-Cantelli lemma.

Theorem 2.3.2 $X_n \rightarrow X$ in probability if and only if for every subsequence $X_{n(m)}$ there is a further subsequence $X_{n(m_k)}$ that converges almost surely to X .

Proof Let ϵ_k be a sequence of positive numbers that $\downarrow 0$. For each k , there is an $n(m_k) > n(m_{k-1})$ so that $P(|X_{n(m_k)} - X| > \epsilon_k) \leq 2^{-k}$. Since

$$\sum_{k=1}^{\infty} P(|X_{n(m_k)} - X| > \epsilon_k) < \infty$$

the Borel-Cantelli lemma implies $P(|X_{n(m_k)} - X| > \epsilon_k \text{ i.o.}) = 0$, i.e., $X_{n(m_k)} \rightarrow X$ a.s. To prove the second conclusion, we note that if for every subsequence $X_{n(m)}$ there is a further subsequence $X_{n(m_k)}$ that converges almost surely to X , then we can apply the next lemma to the sequence of numbers $y_n = P(|X_n - X| > \delta)$ for any $\delta > 0$ to get the desired result. $\square$

Theorem 2.3.3 *Let y_n be a sequence of elements of a topological space. If every subsequence $y_{n(m)}$ has a further subsequence $y_{n(m_k)}$ that converges to y , then $y_n \rightarrow y$.*

Proof If $y_n \not\rightarrow y$, then there is an open set G containing y and a subsequence $y_{n(m)}$ with $y_{n(m)} \notin G$ for all m , but clearly no subsequence of $y_{n(m)}$ converges to y . $\square$

Remark Since there is a sequence of random variables that converges in probability but not a.s. (for an example, see Exercises 2.3.14 or 2.3.11), it follows from Theorem 2.3.3 that a.s. convergence does not come from a metric, or even from a topology. Exercise 2.3.6 will give a metric for convergence in probability, and Exercise 2.3.7 will show that the space of random variables is a complete space under this metric.

Theorem 2.3.2 allows us to upgrade convergence in probability to convergence almost surely. An example of the usefulness of this is:

Theorem 2.3.4 *If f is continuous and $X_n \rightarrow X$ in probability, then $f(X_n) \rightarrow f(X)$ in probability. If, in addition, f is bounded, then $Ef(X_n) \rightarrow Ef(X)$.*

Proof If $X_{n(m)}$ is a subsequence, then Theorem 2.3.2 implies there is a further subsequence $X_{n(m_k)} \rightarrow X$ almost surely. Since f is continuous, Exercise 1.3.3 implies $f(X_{n(m_k)}) \rightarrow f(X)$ almost surely and Theorem 2.3.2 implies $f(X_n) \rightarrow f(X)$ in probability. If f is bounded, then the bounded convergence theorem implies $Ef(X_{n(m_k)}) \rightarrow Ef(X)$, and applying Theorem 2.3.3 to $y_n = Ef(X_n)$ gives the desired result. $\square$

Exercise 2.3.2 Prove the first result in Theorem 2.3.4 directly from the definition.

As our second application of the Borel-Cantelli lemma, we get our first strong law of large numbers:

Theorem 2.3.5 *Let $X_1, X_2, \dots$ be i.i.d. with $EX_i = \mu$ and $EX_i^4 < \infty$. If $S_n = X_1 + \dots + X_n$, then $S_n/n \rightarrow \mu$ a.s.*

Proof By letting $X'_i = X_i - \mu$, we can suppose without loss of generality that $\mu = 0$. Now

$$ES_n^4 = E \left(\sum_{i=1}^n X_i \right)^4 = E \sum_{1 \leq i, j, k, \ell \leq n} X_i X_j X_k X_\ell$$

Terms in the sum of the form $E(X_i^3 X_j)$, $E(X_i^2 X_j X_k)$, and $E(X_i X_j X_k X_\ell)$ are 0 (if i, j, k, ℓ are distinct) since the expectation of the product is the product of the expectations, and in

each case one of the terms has expectation 0. The only terms that do not vanish are those of the form EX_i^4 and $EX_i^2 X_j^2 = (EX_i^2)^2$. There are n and $3n(n-1)$ of these terms, respectively. (In the second case, we can pick the two indices in $n(n-1)/2$ ways, and with the indices fixed, the term can arise in a total of six ways.) The last observation implies

$$ES_n^4 = nEX_1^4 + 3(n^2 - n)(EX_1^2)^2 \leq Cn^2$$

where $C < \infty$. Chebyshev's inequality gives us

$$P(|S_n| > n\epsilon) \leq E(S_n^4)/(n\epsilon)^4 \leq C/(n^2\epsilon^4)$$

Summing on n and using the Borel-Cantelli lemma gives $P(|S_n| > n\epsilon \text{ i.o.}) = 0$. Since ϵ is arbitrary, the proof is complete. $\square$

The converse of the Borel-Cantelli lemma is trivially false.

Example 2.3.6 Let $\Omega = (0, 1)$, $\mathcal{F}$ = Borel sets, P = Lebesgue measure. If $A_n = (0, a_n)$ where $a_n \rightarrow 0$ as $n \rightarrow \infty$, then $\limsup A_n = \emptyset$, but if $a_n \geq 1/n$, we have $\sum a_n = \infty$.

The example just given suggests that for general sets we cannot say much more than the result in Exercise 2.3.1.

For independent events, however, the necessary condition for $P(\limsup A_n) > 0$ is sufficient for $P(\limsup A_n) = 1$.

Theorem 2.3.7 (The second Borel-Cantelli lemma) *If the events A_n are independent, then $\sum P(A_n) = \infty$ implies $P(A_n \text{ i.o.}) = 1$.*

Proof Let $M < N < \infty$. Independence and $1 - x \leq e^{-x}$ imply

$$\begin{aligned} P\left(\bigcap_{n=M}^N A_n^c\right) &= \prod_{n=M}^N (1 - P(A_n)) \leq \prod_{n=M}^N \exp(-P(A_n)) \\ &= \exp\left(-\sum_{n=M}^N P(A_n)\right) \rightarrow 0 \quad \text{as } N \rightarrow \infty \end{aligned}$$

So $P(\bigcup_{n=M}^\infty A_n) = 1$ for all M , and since $\bigcup_{n=M}^\infty A_n \downarrow \limsup A_n$ it follows that $P(\limsup A_n) = 1$. $\square$

A typical application of the second Borel-Cantelli lemma is:

Theorem 2.3.8 *If $X_1, X_2, \dots$ are i.i.d. with $E|X_i| = \infty$, then $P(|X_n| \geq n \text{ i.o.}) = 1$. So if $S_n = X_1 + \dots + X_n$, then $P(\lim S_n/n \text{ exists} \in (-\infty, \infty)) = 0$.*

Proof From Lemma 2.2.13, we get

$$E|X_1| = \int_0^\infty P(|X_1| > x) dx \leq \sum_{n=0}^\infty P(|X_1| > n)$$

Since $E|X_1| = \infty$ and $X_1, X_2, \dots$ are i.i.d., it follows from the second Borel-Cantelli lemma that $P(|X_n| \geq n \text{ i.o.}) = 1$. To prove the second claim, observe that

$$\frac{S_n}{n} - \frac{S_{n+1}}{n+1} = \frac{S_n}{n(n+1)} - \frac{X_{n+1}}{n+1}$$

and on $C \equiv \{\omega : \lim_{n \rightarrow \infty} S_n/n \text{ exists} \in (-\infty, \infty)\}$, $S_n/(n(n+1)) \rightarrow 0$. So, on $C \cap \{\omega : |X_n| \geq n \text{ i.o.}\}$, we have

$$\left| \frac{S_n}{n} - \frac{S_{n+1}}{n+1} \right| > 2/3 \quad \text{i.o.}$$

contradicting the fact that $\omega \in C$. From the last observation, we conclude that

$$\{\omega : |X_n| \geq n \text{ i.o.}\} \cap C = \emptyset$$

and since $P(|X_n| \geq n \text{ i.o.}) = 1$, it follows that $P(C) = 0$. $\square$

Theorem 2.3.8 shows that $E|X_i| < \infty$ is necessary for the strong law of large numbers. The reader will have to wait until Theorem 2.4.1 to see that condition is also sufficient. The next result extends the second Borel-Cantelli lemma and sharpens its conclusion.

Theorem 2.3.9 *If $A_1, A_2, \dots$ are pairwise independent and $\sum_{n=1}^{\infty} P(A_n) = \infty$, then as $n \rightarrow \infty$*

$$\sum_{m=1}^n 1_{A_m} \bigg/ \sum_{m=1}^n P(A_m) \rightarrow 1 \quad \text{a.s.}$$

Proof Let $X_m = 1_{A_m}$ and let $S_n = X_1 + \dots + X_n$. Since the A_m are pairwise independent, the X_m are uncorrelated, and hence Theorem 2.2.1 implies

$$\text{var}(S_n) = \text{var}(X_1) + \dots + \text{var}(X_n)$$

$\text{var}(X_m) \leq E(X_m^2) = E(X_m)$, since $X_m \in \{0, 1\}$, so $\text{var}(S_n) \leq E(S_n)$. Chebyshev's inequality implies

$$(*) \quad P(|S_n - ES_n| > \delta ES_n) \leq \text{var}(S_n)/(\delta ES_n)^2 \leq 1/(\delta^2 ES_n) \rightarrow 0$$

as $n \rightarrow \infty$. (Since we have assumed $ES_n \rightarrow \infty$.)

The last computation shows that $S_n/ES_n \rightarrow 1$ in probability. To get almost sure convergence, we have to take subsequences. Let $n_k = \inf\{n : ES_n \geq k^2\}$. Let $T_k = S_{n_k}$ and note that the definition and $EX_m \leq 1$ imply $k^2 \leq ET_k \leq k^2 + 1$. Replacing n by n_k in (*) and using $ET_k \geq k^2$ shows

$$P(|T_k - ET_k| > \delta ET_k) \leq 1/(\delta^2 k^2)$$

So $\sum_{k=1}^{\infty} P(|T_k - ET_k| > \delta ET_k) < \infty$, and the Borel-Cantelli lemma implies $P(|T_k - ET_k| > \delta ET_k \text{ i.o.}) = 0$. Since δ is arbitrary, it follows that $T_k/ET_k \rightarrow 1$ a.s. To show $S_n/ES_n \rightarrow 1$ a.s., pick an ω so that $T_k(\omega)/ET_k \rightarrow 1$ and observe that if $n_k \leq n < n_{k+1}$ then

$$\frac{T_k(\omega)}{ET_{k+1}} \leq \frac{S_n(\omega)}{ES_n} \leq \frac{T_{k+1}(\omega)}{ET_k}$$

To show that the terms at the left and right ends $\rightarrow 1$, we rewrite the last inequalities as

$$\frac{ET_k}{ET_{k+1}} \cdot \frac{T_k(\omega)}{ET_k} \leq \frac{S_n(\omega)}{ES_n} \leq \frac{T_{k+1}(\omega)}{ET_{k+1}} \cdot \frac{ET_{k+1}}{ET_k}$$

From this, we see it is enough to show $ET_{k+1}/ET_k \rightarrow 1$, but this follows from

$$k^2 \leq ET_k \leq ET_{k+1} \leq (k+1)^2 + 1$$

and the fact that $\{(k+1)^2 + 1\}/k^2 = 1 + 2/k + 2/k^2 \rightarrow 1$. $\square$

The moral of the proof of Theorem 2.3.9 is that if you want to show that $X_n/c_n \rightarrow 1$ a.s. for sequences $c_n, X_n \geq 0$ that are increasing, it is enough to prove the result for a subsequence $n(k)$ that has $c_{n(k+1)}/c_{n(k)} \rightarrow 1$.

Example 2.3.10 (Record values) Let $X_1, X_2, \dots$ be a sequence of random variables and think of X_k as the distance for an individual's k th high jump or shot-put toss so that $A_k = \{X_k > \sup_{j < k} X_j\}$ is the event that a record occurs at time k . Ignoring the fact that an athlete's performance may get better with more experience or that injuries may occur, we will suppose that $X_1, X_2, \dots$ are i.i.d. with a distribution $F(x)$ that is continuous. Even though it may seem that the occurrence of a record at time k will make it less likely that one will occur at time $k + 1$, we

Claim. The A_k are independent with $P(A_k) = 1/k$.

To prove this, we start by observing that since F is continuous $P(X_j = X_k) = 0$ for any $j \neq k$ (see Exercise 2.1.5), so we can let $Y_1^n > Y_2^n > \dots > Y_n^n$ be the random variables $X_1, \dots, X_n$ put into decreasing order and define a random permutation of $\{1, \dots, n\}$ by $\pi_n(i) = j$ if $X_i = Y_j^n$, i.e., if the i th random variable has rank j . Since the distribution of $(X_1, \dots, X_n)$ is not affected by changing the order of the random variables, it is easy to see:

(a) The permutation π_n is uniformly distributed over the set of $n!$ possibilities.

Proof of (a) This is “obvious” by symmetry, but if one wants to hear more, we can argue as follows. Let π_n be the permutation induced by $(X_1, \dots, X_n)$, and let σ_n be a randomly chosen permutation of $\{1, \dots, n\}$ independent of the X sequence. Then we can say two things about the permutation induced by $(X_{\sigma(1)}, \dots, X_{\sigma(n)})$: (i) it is $\pi_n \circ \sigma_n$, and (ii) it has the same distribution as π_n . The desired result follows now by noting that if π is any permutation, $\pi \circ \sigma_n$, is uniform over the $n!$ possibilities. $\square$

Once you believe (a), the rest is easy:

(b) $P(A_n) = P(\pi_n(n) = 1) = 1/n$.

(c) If $m < n$ and $i_{m+1}, \dots, i_n$ are distinct elements of $\{1, \dots, n\}$, then

$$P(A_m | \pi_n(j) = i_j \text{ for } m+1 \leq j \leq n) = 1/m$$

Intuitively, this is true since if we condition on the ranks of $X_{m+1}, \dots, X_n$, then this determines the set of ranks available for $X_1, \dots, X_m$, but all possible orderings of the ranks are equally likely, and hence there is probability $1/m$ that the smallest rank will end up at m .

Proof of (c) If we let σ_m be a randomly chosen permutation of $\{1, \dots, m\}$, then (i) $\pi_n \circ \sigma_m$ has the same distribution as π_n , and (ii) since the application of σ_m randomly rearranges $\pi_n(1), \dots, \pi_n(m)$ the desired result follows. $\square$

If we let $m_1 < m_2 < \dots < m_k$, then it follows from (c) that

$$P(A_{m_1} | A_{m_2} \cap \dots \cap A_{m_k}) = P(A_{m_1})$$

and the claim follows by induction.

Using Theorem 2.3.9 and the by now familiar fact that $\sum_{m=1}^n 1/m \sim \log n$, we have:

Theorem 2.3.11 *If $R_n = \sum_{m=1}^n 1_{A_m}$ is the number of records at time n . then as $n \rightarrow \infty$,*

$$R_n / \log n \rightarrow 1 \quad \text{a.s.}$$

The reader should note that the last result is independent of the distribution F (as long as it is continuous).

Remark Let $X_1, X_2, \dots$ be i.i.d. with a distribution that is continuous. Let Y_i be the number of $j \leq i$ with $X_j > X_i$. It follows from (a) that Y_i are independent random variables with $P(Y_i = j) = 1/i$ for $0 \leq j < i - 1$.

Comic relief Let $X_0, X_1, \dots$ be i.i.d. and imagine they are the offers you get for a car you are going to sell. Let $N = \inf\{n \geq 1 : X_n > X_0\}$. Symmetry implies $P(N > n) \geq 1/(n+1)$. (When the distribution is continuous, this probability is exactly $1/(n+1)$, but our distribution now is general and ties go to the first person who calls.) Using Exercise 2.2.7 now:

$$EN = \sum_{n=0}^{\infty} P(N > n) \geq \sum_{n=0}^{\infty} \frac{1}{n+1} = \infty$$

so the expected time you have to wait until you get an offer better than the first one is ∞ . To avoid lawsuits, let me hasten to add that I am not suggesting that you should take the first offer you get!

Example 2.3.12 (Head runs) Let $X_n, n \in \mathbf{Z}$, be i.i.d. with $P(X_n = 1) = P(X_n = -1) = 1/2$. Let $\ell_n = \max\{m : X_{n-m+1} = \dots = X_n = 1\}$ be the length of the run of +1's at time n , and let $L_n = \max_{1 \leq m \leq n} \ell_m$ be the longest run at time n . We use a two-sided sequence so that for all n , $P(\ell_n = k) = (1/2)^{k+1}$ for $k \geq 0$. Since $\ell_1 < \infty$, the result we are going to prove

$$L_n / \log_2 n \rightarrow 1 \quad \text{a.s.} \tag{2.3.1}$$

is also true for a one-sided sequence. To prove (2.3.1), we begin by observing

$$P(\ell_n \geq (1 + \epsilon) \log_2 n) \leq n^{-(1+\epsilon)}$$

for any $\epsilon > 0$, so it follows from the Borel-Cantelli lemma that $\ell_n \leq (1 + \epsilon) \log_2 n$ for $n \geq N_\epsilon$. Since ϵ is arbitrary, it follows that

$$\limsup_{n \rightarrow \infty} L_n / \log_2 n \leq 1 \quad \text{a.s.}$$

To get a result in the other direction, we break the first n trials into disjoint blocks of length $[(1 - \epsilon) \log_2 n] + 1$, on which the variables are all 1 with probability

$$2^{-[(1-\epsilon) \log_2 n] - 1} \geq n^{-(1-\epsilon)}/2,$$

to conclude that if n is large enough so that $[n/[(1 - \epsilon) \log_2 n] + 1]] \geq n / \log_2 n$

$$P(L_n \leq (1 - \epsilon) \log_2 n) \leq (1 - n^{-(1-\epsilon)}/2)^{n/(\log_2 n)} \leq \exp(-n^\epsilon/2 \log_2 n)$$

which is summable, so the Borel-Cantelli lemma implies

$$\liminf_{n \rightarrow \infty} L_n / \log_2 n \geq 1 \quad \text{a.s.}$$

Exercise 2.3.3 Let ℓ_n be the length of the head run at time n . See Example 8.3.1 for the precise definition. Show that $\limsup_{n \rightarrow \infty} \ell_n / \log_2 n = 1$, $\liminf_{n \rightarrow \infty} \ell_n = 0$ a.s.

Exercises

- 2.3.4 **Fatou's lemma.** Suppose $X_n \geq 0$ and $X_n \rightarrow X$ in probability. Show that $\liminf_{n \rightarrow \infty} EX_n \geq EX$.
- 2.3.5 **Dominated convergence.** Suppose $X_n \rightarrow X$ in probability and (a) $|X_n| \leq Y$ with $EY < \infty$ or (b) there is a continuous function g with $g(x) > 0$ for large x with $|x|/g(x) \rightarrow 0$ as $|x| \rightarrow \infty$ so that $Eg(X_n) \leq C < \infty$ for all n . Show that $EX_n \rightarrow EX$.
- 2.3.6 **Metric for convergence in probability.** Show (a) that $d(X, Y) = E(|X - Y|/(1 + |X - Y|))$ defines a metric on the set of random variables, i.e., (i) $d(X, Y) = 0$ if and only if $X = Y$ a.s., (ii) $d(X, Y) = d(Y, X)$, (iii) $d(X, Z) \leq d(X, Y) + d(Y, Z)$ and (b) that $d(X_n, X) \rightarrow 0$ as $n \rightarrow \infty$ if and only if $X_n \rightarrow X$ in probability.
- 2.3.7 Show that random variables are a complete space under the metric defined in the previous exercise, i.e., if $d(X_m, X_n) \rightarrow 0$ whenever $m, n \rightarrow \infty$, then there is a r.v. X_∞ so that $X_n \rightarrow X_\infty$ in probability.
- 2.3.8 Let A_n be a sequence of independent events with $P(A_n) < 1$ for all n . Show that $P(\cup A_n) = 1$ implies $\sum_n P(A_n) = \infty$ and hence $P(A_n \text{ i.o.}) = 1$.
- 2.3.9 (i) If $P(A_n) \rightarrow 0$ and $\sum_{n=1}^\infty P(A_n^c \cap A_{n+1}) < \infty$, then $P(A_n \text{ i.o.}) = 0$. (ii) Find an example of a sequence A_n to which the result in (i) can be applied but the Borel-Cantelli lemma cannot.
- 2.3.10 **Kochen-Stone lemma.** Suppose $\sum P(A_k) = \infty$. Use Exercises 1.6.6 and 2.3.1 to show that if

$$\limsup_{n \rightarrow \infty} \left(\sum_{k=1}^n P(A_k) \right)^2 / \left(\sum_{1 \leq j, k \leq n} P(A_j \cap A_k) \right) = \alpha > 0$$

then $P(A_n \text{ i.o.}) \geq \alpha$. The case $\alpha = 1$ contains Theorem 2.3.7.

- 2.3.11 Let $X_1, X_2, \dots$ be independent with $P(X_n = 1) = p_n$ and $P(X_n = 0) = 1 - p_n$. Show that (i) $X_n \rightarrow 0$ in probability if and only if $p_n \rightarrow 0$, and (ii) $X_n \rightarrow 0$ a.s. if and only if $\sum p_n < \infty$.
- 2.3.12 Let $X_1, X_2, \dots$ be a sequence of r.v.'s on $(\Omega, \mathcal{F}, P)$, where Ω is a countable set and $\mathcal{F}$ consists of all subsets of Ω . Show that $X_n \rightarrow X$ in probability implies $X_n \rightarrow X$ a.s.
- 2.3.13 If X_n is any sequence of random variables, there are constants $c_n \rightarrow \infty$ so that $X_n/c_n \rightarrow 0$ a.s.
- 2.3.14 Let $X_1, X_2, \dots$ be independent. Show that $\sup X_n < \infty$ a.s. if and only if $\sum_n P(X_n > A) < \infty$ for some A .

- 2.3.15 Let $X_1, X_2, \dots$ be i.i.d. with $P(X_i > x) = e^{-x}$, let $M_n = \max_{1 \leq m \leq n} X_m$. Show that
 (i) $\limsup_{n \rightarrow \infty} X_n / \log n = 1$ a.s. and (ii) $M_n / \log n \rightarrow 1$ a.s.
- 2.3.16 Let $X_1, X_2, \dots$ be i.i.d. with distribution F , let $\lambda_n \uparrow \infty$, and let $A_n = \{\max_{1 \leq m \leq n} X_m > \lambda_n\}$. Show that $P(A_n \text{ i.o.}) = 0$ or 1 according as $\sum_{n \geq 1} (1 - F(\lambda_n)) < \infty$ or $= \infty$.
- 2.3.17 Let $Y_1, Y_2, \dots$ be i.i.d. Find necessary and sufficient conditions for
 (i) $Y_n/n \rightarrow 0$ almost surely, (ii) $(\max_{m \leq n} Y_m)/n \rightarrow 0$ almost surely,
 (iii) $(\max_{m \leq n} Y_m)/n \rightarrow 0$ in probability, and (iv) $Y_n/n \rightarrow 0$ in probability.
- 2.3.18 Let $0 \leq X_1 \leq X_2 \leq \dots$ be random variables with $EX_n \sim an^\alpha$ with $a, \alpha > 0$, and $\text{var}(X_n) \leq Bn^\beta$ with $\beta < 2\alpha$. Show that $X_n/n^\alpha \rightarrow a$ a.s.
- 2.3.19 Let X_n be independent Poisson r.v.'s with $EX_n = \lambda_n$, and let $S_n = X_1 + \dots + X_n$. Show that if $\sum \lambda_n = \infty$, then $S_n/ES_n \rightarrow 1$ a.s.
- 2.3.20 Show that if X_n is the outcome of the n th play of the St. Petersburg game (Example 2.2.16), then $\limsup_{n \rightarrow \infty} X_n/(n \log_2 n) = \infty$ a.s. and hence the same result holds for S_n . This shows that the convergence $S_n/(n \log_2 n) \rightarrow 1$ in probability proved in Section 2.2 does not occur a.s.

2.4 Strong Law of Large Numbers

We are now ready to give Etemadi's (1981) proof of:

Theorem 2.4.1 (Strong law of large numbers) *Let $X_1, X_2, \dots$ be pairwise independent identically distributed random variables with $E|X_i| < \infty$. Let $EX_i = \mu$ and $S_n = X_1 + \dots + X_n$. Then $S_n/n \rightarrow \mu$ a.s. as $n \rightarrow \infty$.*

Proof As in the proof of the weak law of large numbers, we begin by truncating.

Lemma 2.4.2 *Let $Y_k = X_k 1_{(|X_k| \leq k)}$ and $T_n = Y_1 + \dots + Y_n$. It is sufficient to prove that $T_n/n \rightarrow \mu$ a.s.*

Proof $\sum_{k=1}^{\infty} P(|X_k| > k) \leq \int_0^{\infty} P(|X_1| > t) dt = E|X_1| < \infty$ so $P(X_k \neq Y_k \text{ i.o.}) = 0$. This shows that $|S_n(\omega) - T_n(\omega)| \leq R(\omega) < \infty$ a.s. for all n , from which the desired result follows. $\square$

The second step is not so intuitive, but it is an important part of this proof and the one given in Section 2.5.

Lemma 2.4.3 $\sum_{k=1}^{\infty} \text{var}(Y_k)/k^2 \leq 4E|X_1| < \infty$.

Proof To bound the sum, we observe

$$\text{var}(Y_k) \leq E(Y_k^2) = \int_0^{\infty} 2yP(|Y_k| > y) dy \leq \int_0^k 2yP(|X_1| > y) dy$$

so using Fubini's theorem (since everything is ≥ 0 and the sum is just an integral with respect to counting measure on $\{1, 2, \dots\}$)

$$\begin{aligned} \sum_{k=1}^{\infty} E(Y_k^2)/k^2 &\leq \sum_{k=1}^{\infty} k^{-2} \int_0^{\infty} 1_{(y < k)} 2y P(|X_1| > y) dy \\ &= \int_0^{\infty} \left\{ \sum_{k=1}^{\infty} k^{-2} 1_{(y < k)} \right\} 2y P(|X_1| > y) dy \end{aligned}$$

Since $E|X_1| = \int_0^{\infty} P(|X_1| > y) dy$, we can complete the proof by showing:

Lemma 2.4.4 *If $y \geq 0$ then $2y \sum_{k > y} k^{-2} \leq 4$.*

Proof We begin with the observation that if $m \geq 2$, then

$$\sum_{k \geq m} k^{-2} \leq \int_{m-1}^{\infty} x^{-2} dx = (m-1)^{-1}$$

When $y \geq 1$, the sum starts with $k = [y] + 1 \geq 2$, so

$$2y \sum_{k > y} k^{-2} \leq 2y/[y] \leq 4$$

since $y/[y] \leq 2$ for $y \geq 1$ (the worst case being y close to 2). To cover $0 \leq y < 1$, we note that in this case

$$2y \sum_{k > y} k^{-2} \leq 2 \left(1 + \sum_{k=2}^{\infty} k^{-2} \right) \leq 4$$

This establishes Lemma 2.4.4, which completes the proof of Lemma 2.4.3 and of the theorem. $\square$

The first two steps, Lemmas 2.4.2 and 2.4.3, are standard. Etemadi's inspiration was that since $X_n^+, n \geq 1$, and $X_n^-, n \geq 1$ satisfy the assumptions of the theorem and $X_n = X_n^+ - X_n^-$, we can without loss of generality suppose $X_n \geq 0$. As in the proof of Theorem 2.3.9, we will prove the result first for a subsequence and then use monotonicity to control the values in between. This time, however, we let $\alpha > 1$ and $k(n) = [\alpha^n]$. Chebyshev's inequality implies that if $\epsilon > 0$

$$\begin{aligned} \sum_{n=1}^{\infty} P(|T_{k(n)} - ET_{k(n)}| > \epsilon k(n)) &\leq \epsilon^{-2} \sum_{n=1}^{\infty} \text{var}(T_{k(n)})/k(n)^2 \\ &= \epsilon^{-2} \sum_{n=1}^{\infty} k(n)^{-2} \sum_{m=1}^{k(n)} \text{var}(Y_m) = \epsilon^{-2} \sum_{m=1}^{\infty} \text{var}(Y_m) \sum_{n: k(n) \geq m} k(n)^{-2} \end{aligned}$$

where we have used Fubini's theorem to interchange the two summations of nonnegative terms. Now $k(n) = [\alpha^n]$ and $[\alpha^n] \geq \alpha^n/2$ for $n \geq 1$, so summing the geometric series and noting that the first term is $\leq m^{-2}$:

$$\sum_{n: \alpha^n \geq m} [\alpha^n]^{-2} \leq 4 \sum_{n: \alpha^n \geq m} \alpha^{-2n} \leq 4(1 - \alpha^{-2})^{-1} m^{-2}$$

Combining our computations shows

$$\sum_{n=1}^{\infty} P(|T_{k(n)} - ET_{k(n)}| > \epsilon k(n)) \leq 4(1 - \alpha^{-2})^{-1} \epsilon^{-2} \sum_{m=1}^{\infty} E(Y_m^2) m^{-2} < \infty$$

by Lemma 2.4.3. Since ϵ is arbitrary $(T_{k(n)} - ET_{k(n)})/k(n) \rightarrow 0$ a.s. The dominated convergence theorem implies $EY_k \rightarrow EX_1$ as $k \rightarrow \infty$, so $ET_{k(n)}/k(n) \rightarrow EX_1$ and we have shown $T_{k(n)}/k(n) \rightarrow EX_1$ a.s. To handle the intermediate values, we observe that if $k(n) \leq m < k(n+1)$

$$\frac{T_{k(n)}}{k(n+1)} \leq \frac{T_m}{m} \leq \frac{T_{k(n+1)}}{k(n)}$$

(here we use $Y_i \geq 0$), so recalling $k(n) = [\alpha^n]$, we have $k(n+1)/k(n) \rightarrow \alpha$ and

$$\frac{1}{\alpha} EX_1 \leq \liminf_{n \rightarrow \infty} T_m/m \leq \limsup_{m \rightarrow \infty} T_m/m \leq \alpha EX_1$$

Since $\alpha > 1$ is arbitrary, the proof is complete. $\square$

The next result shows that the strong law holds whenever EX_i exists.

Theorem 2.4.5 *Let $X_1, X_2, \dots$ be i.i.d. with $EX_i^+ = \infty$ and $EX_i^- < \infty$. If $S_n = X_1 + \dots + X_n$, then $S_n/n \rightarrow \infty$ a.s.*

Proof Let $M > 0$ and $X_i^M = X_i \wedge M$. The X_i^M are i.i.d. with $E|X_i^M| < \infty$, so if $S_n^M = X_1^M + \dots + X_n^M$, then Theorem 2.4.1 implies $S_n^M/n \rightarrow EX_i^M$. Since $X_i \geq X_i^M$, it follows that

$$\liminf_{n \rightarrow \infty} S_n/n \geq \lim_{n \rightarrow \infty} S_n^M/n = EX_i^M$$

The monotone convergence theorem implies $E(X_i^M)^+ \uparrow EX_i^+ = \infty$ as $M \uparrow \infty$, so $EX_i^M = E(X_i^M)^+ - E(X_i^M)^- \uparrow \infty$, and we have $\liminf_{n \rightarrow \infty} S_n/n \geq \infty$, which implies the desired result. $\square$

The rest of this section is devoted to applications of the strong law of large numbers.

Example 2.4.6 (Renewal theory) Let $X_1, X_2, \dots$ be i.i.d. with $0 < X_i < \infty$. Let $T_n = X_1 + \dots + X_n$ and think of T_n as the time of n th occurrence of some event. For a concrete situation, consider a diligent janitor who replaces a light bulb the instant it burns out. Suppose the first bulb is put in at time 0 and let X_i be the lifetime of the i th light bulb. In this interpretation, T_n is the time the n th light bulb burns out and $N_t = \sup\{n : T_n \leq t\}$ is the number of light bulbs that have burnt out by time t .

Theorem 2.4.7 *If $EX_1 = \mu \leq \infty$, then as $t \rightarrow \infty$,*

$$N_t/t \rightarrow 1/\mu \text{ a.s. } (1/\infty = 0).$$

Proof By Theorems 2.4.1 and 2.4.5, $T_n/n \rightarrow \mu$ a.s. From the definition of N_t , it follows that $T(N_t) \leq t < T(N_t + 1)$, so dividing through by N_t gives

$$\frac{T(N_t)}{N_t} \leq \frac{t}{N_t} \leq \frac{T(N_t + 1)}{N_t + 1} \cdot \frac{N_t + 1}{N_t}$$

To take the limit, we note that since $T_n < \infty$ for all n , we have $N_t \uparrow \infty$ as $t \rightarrow \infty$. The strong law of large numbers implies that for $\omega \in \Omega_0$ with $P(\Omega_0) = 1$, we have $T_n(\omega)/n \rightarrow \mu$, $N_t(\omega) \uparrow \infty$, and hence

$$\frac{T_{N_t(\omega)}(\omega)}{N_t(\omega)} \rightarrow \mu \quad \frac{N_t(\omega) + 1}{N_t(\omega)} \rightarrow 1$$

From this it follows that for $\omega \in \Omega_0$ that $t/N_t(\omega) \rightarrow \mu$ a.s. □

The last argument shows that if $X_n \rightarrow X_\infty$ a.s. and $N(n) \rightarrow \infty$ a.s., then $X_{N(n)} \rightarrow X_\infty$ a.s. We have written this out with care because the analogous result for convergence in probability is false. If $X_n \in \{0, 1\}$ are independent with $P(X = 1) = a_n \rightarrow 0$ and $\sum_n a_n = \infty$, then $X_n \rightarrow 0$ in probability, but if we let $N(n) = \inf\{m \geq n : X_m = 1\}$, then $X_{N(n)} = 1$ a.s.

Example 2.4.8 (Empirical distribution functions) Let $X_1, X_2, \dots$ be i.i.d. with distribution F and let

$$F_n(x) = n^{-1} \sum_{m=1}^n 1_{(X_m \leq x)}$$

$F_n(x)$ = the observed frequency of values that are $\leq x$, hence the name given here. The next result shows that F_n converges uniformly to F as $n \rightarrow \infty$.

Theorem 2.4.9 (The Glivenko-Cantelli theorem) As $n \rightarrow \infty$,

$$\sup_x |F_n(x) - F(x)| \rightarrow 0 \quad \text{a.s.}$$

Proof Fix x and let $Y_n = 1_{(X_n \leq x)}$. Since the Y_n are i.i.d. with $EY_n = P(X_n \leq x) = F(x)$, the strong law of large numbers implies that $F_n(x) = n^{-1} \sum_{m=1}^n Y_m \rightarrow F(x)$ a.s. In general, if F_n is a sequence of nondecreasing functions that converges pointwise to a bounded and continuous limit F , then $\sup_x |F_n(x) - F(x)| \rightarrow 0$. However, the distribution function $F(x)$ may have jumps, so we have to work a little harder.

Again, fix x and let $Z_n = 1_{(X_n < x)}$. Since the Z_n are i.i.d. with $EZ_n = P(X_n < x) = F(x-) = \lim_{y \uparrow x} F(y)$, the strong law of large numbers implies that $F_n(x-) = n^{-1} \sum_{m=1}^n Z_m \rightarrow F(x-)$ a.s. For $1 \leq j \leq k-1$ let $x_{j,k} = \inf\{y : F(y) \geq j/k\}$. The pointwise convergence of $F_n(x)$ and $F_n(x-)$ imply that we can pick $N_k(\omega)$ so that if $n \geq N_k(\omega)$ then

$$|F_n(x_{j,k}) - F(x_{j,k})| < k^{-1} \quad \text{and} \quad |F_n(x_{j,k}-) - F(x_{j,k}-)| < k^{-1}$$

for $1 \leq j \leq k-1$. If we let $x_{0,k} = -\infty$ and $x_{k,k} = \infty$, then the last two inequalities hold for $j = 0$ or k . If $x \in (x_{j-1,k}, x_{j,k})$ with $1 \leq j \leq k$ and $n \geq N_k(\omega)$, then using the monotonicity of F_n and F , and $F(x_{j,k}-) - F(x_{j-1,k}) \leq k^{-1}$, we have

$$\begin{aligned} F_n(x) &\leq F_n(x_{j,k}-) \leq F(x_{j,k}-) + k^{-1} \leq F(x_{j-1,k}) + 2k^{-1} \leq F(x) + 2k^{-1} \\ F_n(x) &\geq F_n(x_{j-1,k}) \geq F(x_{j-1,k}) - k^{-1} \geq F(x_{j,k}-) - 2k^{-1} \geq F(x) - 2k^{-1} \end{aligned}$$

so $\sup_x |F_n(x) - F(x)| \leq 2k^{-1}$, and we have proved the result. □

Example 2.4.10 (Shannon's theorem) Let $X_1, X_2, \dots \in \{1, \dots, r\}$ be independent with $P(X_i = k) = p(k) > 0$ for $1 \leq k \leq r$. Here we are thinking of $1, \dots, r$ as the letters of an alphabet, and $X_1, X_2, \dots$ are the successive letters produced by an information source. In this i.i.d. case, it is the proverbial monkey at a typewriter. Let $\pi_n(\omega) = p(X_1(\omega)) \cdots p(X_n(\omega))$ be the probability of the realization we observed in the first n trials. Since $\log \pi_n(\omega)$ is a sum of independent random variables, it follows from the strong law of large numbers that

$$-n^{-1} \log \pi_n(\omega) \rightarrow H \equiv - \sum_{k=1}^r p(k) \log p(k) \text{ a.s.}$$

The constant H is called the **entropy** of the source and is a measure of how random it is. The last result is the **asymptotic equipartition property**: If $\epsilon > 0$, then as $n \rightarrow \infty$

$$P \{ \exp(-n(H + \epsilon)) \leq \pi_n(\omega) \leq \exp(-n(H - \epsilon)) \} \rightarrow 1$$

Exercises

- 2.4.1 **Lazy janitor.** Suppose the i th light bulb burns for an amount of time X_i and then remains burned out for time Y_i before being replaced. Suppose the X_i, Y_i are positive and independent with the X 's having distribution F and the Y 's having distribution G , both of which have finite mean. Let R_t be the amount of time in $[0, t]$ that we have a working light bulb. Show that $R_t/t \rightarrow EX_i/(EX_i + EY_i)$ almost surely.
- 2.4.2 Let $X_0 = (1, 0)$ and define $X_n \in \mathbf{R}^2$ inductively by declaring that X_{n+1} is chosen at random from the ball of radius $|X_n|$ centered at the origin, i.e., $X_{n+1}/|X_n|$ is uniformly distributed on the ball of radius 1 and independent of $X_1, \dots, X_n$. Prove that $n^{-1} \log |X_n| \rightarrow c$ a.s. and compute c .
- 2.4.3 **Investment problem.** We assume that at the beginning of each year you can buy bonds for \$1 that are worth \$ a at the end of the year or stocks that are worth a random amount $V \geq 0$. If you always invest a fixed proportion p of your wealth in bonds, then your wealth at the end of year $n + 1$ is $W_{n+1} = (ap + (1 - p)V_n)W_n$. Suppose $V_1, V_2, \dots$ are i.i.d. with $EV_n^2 < \infty$ and $E(V_n^{-2}) < \infty$. (i) Show that $n^{-1} \log W_n \rightarrow c(p)$ a.s. (ii) Show that $c(p)$ is concave. [Use Theorem A.5.1 in the Appendix to justify differentiating under the expected value.] (iii) By investigating $c'(0)$ and $c'(1)$, give conditions on V that guarantee that the optimal choice of p is in $(0, 1)$. (iv) Suppose $P(V = 1) = P(V = 4) = 1/2$. Find the optimal p as a function of a .

2.5 Convergence of Random Series*

In this section, we will pursue a second approach to the strong law of large numbers based on the convergence of random series. This approach has the advantage that it leads to estimates on the rate of convergence under moment assumptions, Theorems 2.5.11 and 2.5.12, and to a negative result for the infinite mean case, Theorem 2.5.13, which is stronger than the one in Theorem 2.3.8. The first two results in this section are of considerable interest in their own right, although we will see more general versions in Lemma 3.1.1 and Theorem 3.4.2.

To state the first result, we need some notation. Let $\mathcal{F}'_n = \sigma(X_n, X_{n+1}, \dots)$ = the future after time n = the smallest σ -field with respect to which all the $X_m, m \geq n$ are measurable. Let $\mathcal{T} = \cap_n \mathcal{F}'_n$ = the remote future, or **tail σ -field**. Intuitively, $A \in \mathcal{T}$ if and only if changing a finite number of values does not affect the occurrence of the event. As usual, we turn to examples to help explain the definition.

Example 2.5.1 If $B_n \in \mathcal{R}$, then $\{X_n \in B_n \text{ i.o.}\} \in \mathcal{T}$. If we let $X_n = 1_{A_n}$ and $B_n = \{1\}$, this example becomes $\{A_n \text{ i.o.}\}$.

Example 2.5.2 Let $S_n = X_1 + \dots + X_n$. It is easy to check that

$$\begin{aligned} \{\lim_{n \rightarrow \infty} S_n \text{ exists}\} &\in \mathcal{T}, \\ \{\limsup_{n \rightarrow \infty} S_n > 0\} &\notin \mathcal{T}, \\ \{\limsup_{n \rightarrow \infty} S_n/c_n > x\} &\in \mathcal{T} \text{ if } c_n \rightarrow \infty. \end{aligned}$$

The next result shows that all examples are trivial.

Theorem 2.5.3 (Kolmogorov's 0-1 law) *If $X_1, X_2, \dots$ are independent and $A \in \mathcal{T}$, then $P(A) = 0$ or 1 .*

Proof We will show that A is independent of itself, that is, $P(A \cap A) = P(A)P(A)$, so $P(A) = P(A)^2$, and hence $P(A) = 0$ or 1 . We will sneak up on this conclusion in two steps:

(a) $A \in \sigma(X_1, \dots, X_k)$ and $B \in \sigma(X_{k+1}, X_{k+2}, \dots)$ are independent.

Proof of (a). If $B \in \sigma(X_{k+1}, \dots, X_{k+j})$ for some j , this follows from Theorem 2.1.9. Since $\sigma(X_1, \dots, X_k)$ and $\cup_j \sigma(X_{k+1}, \dots, X_{k+j})$ are π -systems that contain Ω (a) follows from Theorem 2.1.7.

(b) $A \in \sigma(X_1, X_2, \dots)$ and $B \in \mathcal{T}$ are independent.

Proof of (b). Since $\mathcal{T} \subset \sigma(X_{k+1}, X_{k+2}, \dots)$, if $A \in \sigma(X_1, \dots, X_k)$ for some k , this follows from (a). $\cup_k \sigma(X_1, \dots, X_k)$ and $\mathcal{T}$ are π -systems that contain Ω , so (b) follows from Theorem 2.1.7.

Since $\mathcal{T} \subset \sigma(X_1, X_2, \dots)$, (b) implies an $A \in \mathcal{T}$ is independent of itself and Theorem 2.5.3 follows. $\square$

Before taking up our main topic, we will prove a 0-1 law that, in the i.i.d. case, generalizes Kolmogorov's. To state the new 0-1 law, we need two definitions. A **finite permutation** of $\mathbf{N} = \{1, 2, \dots\}$ is a map π from $\mathbf{N}$ onto $\mathbf{N}$ so that $\pi(i) \neq i$ for only finitely many i . If π is a finite permutation of $\mathbf{N}$ and $\omega \in S^{\mathbf{N}}$, we define $(\pi\omega)_i = \omega_{\pi(i)}$. In words, the coordinates of ω are rearranged according to π . Since $X_i(\omega) = \omega_i$ this is the same as rearranging the random variables. An event A is **permutable** if $\pi^{-1}A \equiv \{\omega : \pi\omega \in A\}$ is equal to A for any finite permutation π , or, in other words, if its occurrence is not affected by rearranging finitely many of the random variables. The collection of permutable events is a σ -field. It is called the **exchangeable** σ -field and denoted by $\mathcal{E}$.

To see the reason for interest in permutable events, suppose $S = \mathbf{R}$ and let $S_n(\omega) = X_1(\omega) + \dots + X_n(\omega)$. Two examples of permutable events are

- (i) $\{\omega : S_n(\omega) \in B \text{ i.o.}\}$
- (ii) $\{\omega : \limsup_{n \rightarrow \infty} S_n(\omega)/c_n \geq 1\}$

In each case, the event is permutable because $S_n(\omega) = S_n(\pi\omega)$ for large n . The list of examples can be enlarged considerably by observing:

(iii) All events in the tail σ -field $\mathcal{T}$ are permutable.

To see this, observe that if $A \in \sigma(X_{n+1}, X_{n+2}, \dots)$, then the occurrence of A is unaffected by a permutation of $X_1, \dots, X_n$. (i) shows that the converse of (iii) is false. The next result shows that for an i.i.d. sequence there is no difference between $\mathcal{E}$ and $\mathcal{T}$. They are both trivial.

Theorem 2.5.4 (Hewitt-Savage 0-1 law) *If $X_1, X_2, \dots$ are i.i.d. and $A \in \mathcal{E}$ then $P(A) \in \{0, 1\}$.*

Proof Let $A \in \mathcal{E}$. As in the proof of Kolmogorov's 0-1 law, we will show A is independent of itself, i.e., $P(A) = P(A \cap A) = P(A)P(A)$ so $P(A) \in \{0, 1\}$. Let $A_n \in \sigma(X_1, \dots, X_n)$ so that

$$(a) \quad P(A_n \Delta A) \rightarrow 0$$

Here $A \Delta B = (A - B) \cup (B - A)$ is the symmetric difference. The existence of the A_n 's is proved in part ii of Lemma A.2.1. A_n can be written as $\{\omega : (\omega_1, \dots, \omega_n) \in B_n\}$ with $B_n \in \mathcal{S}^n$. Let

$$\pi(j) = \begin{cases} j + n & \text{if } 1 \leq j \leq n \\ j - n & \text{if } n + 1 \leq j \leq 2n \\ j & \text{if } j \geq 2n + 1 \end{cases}$$

Observing that π^2 is the identity (so we don't have to worry about whether to write π or π^{-1}) and the coordinates are i.i.d. (so the permuted coordinates are) gives

$$(b) \quad P(\omega : \omega \in A_n \Delta A) = P(\omega : \pi\omega \in A_n \Delta A)$$

Now $\{\omega : \pi\omega \in A\} = \{\omega : \omega \in A\}$, since A is permutable, and

$$\{\omega : \pi\omega \in A_n\} = \{\omega : (\omega_{n+1}, \dots, \omega_{2n}) \in B_n\}$$

If we use A'_n to denote the last event, then we have

$$(c) \quad \{\omega : \pi\omega \in A_n \Delta A\} = \{\omega : \omega \in A'_n \Delta A\}$$

Combining (b) and (c) gives

$$(d) \quad P(A_n \Delta A) = P(A'_n \Delta A)$$

It is easy to see that

$$|P(B) - P(C)| \leq |P(B \Delta C)|$$

so (d) implies $P(A_n), P(A'_n) \rightarrow P(A)$. Now $A - C \subset (A - B) \cup (B - C)$ and with a similar inequality for $C - A$ implies $A \Delta C \subset (A \Delta B) \cup (B \Delta C)$. The last inequality, (d), and (a) imply

$$P(A_n \Delta A'_n) \leq P(A_n \Delta A) + P(A \Delta A'_n) \rightarrow 0$$

The last result implies

$$\begin{aligned} 0 &\leq P(A_n) - P(A_n \cap A'_n) \\ &\leq P(A_n \cup A'_n) - P(A_n \cap A'_n) = P(A_n \Delta A'_n) \rightarrow 0 \end{aligned}$$

so $P(A_n \cap A'_n) \rightarrow P(A)$. But A_n and A'_n are independent, so

$$P(A_n \cap A'_n) = P(A_n)P(A'_n) \rightarrow P(A)^2$$

This shows $P(A) = P(A)^2$, and proves Theorem 2.5.4. $\square$

If $A_1, A_2, \dots$ are independent, then Theorem 2.5.3 implies $P(A_n \text{ i.o.}) = 0$ or 1. Applying Theorem 2.5.3 to Example 2.5.2 gives $P(\lim_{n \rightarrow \infty} S_n \text{ exists}) = 0$ or 1. The next result will help us prove the probability is 1 in certain situations.

Theorem 2.5.5 (Kolmogorov's maximal inequality) Suppose $X_1, \dots, X_n$ are independent with $EX_i = 0$ and $\text{var}(X_i) < \infty$. If $S_n = X_1 + \dots + X_n$, then

$$P\left(\max_{1 \leq k \leq n} |S_k| \geq x\right) \leq x^{-2} \text{var}(S_n)$$

Remark Under the same hypotheses, Chebyshev's inequality (Theorem 1.6.4) gives only

$$P(|S_n| \geq x) \leq x^{-2} \text{var}(S_n)$$

Proof Let $A_k = \{|S_k| \geq x \text{ but } |S_j| < x \text{ for } j < k\}$, i.e., we break things down according to the time that $|S_k|$ first exceeds x . Since the A_k are disjoint and $(S_n - S_k)^2 \geq 0$,

$$\begin{aligned} ES_n^2 &\geq \sum_{k=1}^n \int_{A_k} S_n^2 dP = \sum_{k=1}^n \int_{A_k} S_k^2 + 2S_k(S_n - S_k) + (S_n - S_k)^2 dP \\ &\geq \sum_{k=1}^n \int_{A_k} S_k^2 dP + \sum_{k=1}^n \int_{A_k} 2S_k 1_{A_k} \cdot (S_n - S_k) dP \end{aligned}$$

$S_k 1_{A_k} \in \sigma(X_1, \dots, X_k)$ and $S_n - S_k \in \sigma(X_{k+1}, \dots, X_n)$ are independent by Theorem 2.1.10, so using Theorem 2.1.13 and $E(S_n - S_k) = 0$ shows

$$\int 2S_k 1_{A_k} \cdot (S_n - S_k) dP = E(2S_k 1_{A_k}) \cdot E(S_n - S_k) = 0$$

Using now the fact that $|S_k| \geq x$ on A_k and the A_k are disjoint,

$$ES_n^2 \geq \sum_{k=1}^n \int_{A_k} S_k^2 dP \geq \sum_{k=1}^n x^2 P(A_k) = x^2 P\left(\max_{1 \leq k \leq n} |S_k| \geq x\right) \quad \square$$

We turn now to our results on convergence of series. To state them, we need a definition. We say that $\sum_{n=1}^{\infty} a_n$ converges if $\lim_{N \rightarrow \infty} \sum_{n=1}^N a_n$ exists.

Theorem 2.5.6 Suppose $X_1, X_2, \dots$ are independent and have $EX_n = 0$. If

$$\sum_{n=1}^{\infty} \text{var}(X_n) < \infty$$

then with probability one $\sum_{n=1}^{\infty} X_n(\omega)$ converges.

Proof Let $S_N = \sum_{n=1}^N X_n$. From Theorem 2.5.5, we get

$$P\left(\max_{M \leq m \leq N} |S_m - S_M| > \epsilon\right) \leq \epsilon^{-2} \text{var}(S_N - S_M) = \epsilon^{-2} \sum_{n=M+1}^N \text{var}(X_n)$$

Letting $N \rightarrow \infty$ in the last result, we get

$$P\left(\sup_{m \geq M} |S_m - S_M| > \epsilon\right) \leq \epsilon^{-2} \sum_{n=M+1}^{\infty} \text{var}(X_n) \rightarrow 0 \quad \text{as } M \rightarrow \infty$$

If we let $w_M = \sup_{m, n \geq M} |S_m - S_n|$, then $w_M \downarrow$ as $M \uparrow$ and

$$P(w_M > 2\epsilon) \leq P\left(\sup_{m \geq M} |S_m - S_M| > \epsilon\right) \rightarrow 0$$

as $M \rightarrow \infty$ so $w_M \downarrow 0$ almost surely. But $w_M(\omega) \downarrow 0$ implies $S_n(\omega)$ is a Cauchy sequence and hence $\lim_{n \rightarrow \infty} S_n(\omega)$ exists, so the proof is complete. $\square$

Example 2.5.7 Let $X_1, X_2, \dots$ be independent with

$$P(X_n = n^{-\alpha}) = P(X_n = -n^{-\alpha}) = 1/2$$

$EX_n = 0$ and $\text{var}(X_n) = n^{-2\alpha}$ so if $\alpha > 1/2$, it follows from Theorem 2.5.6 that $\sum X_n$ converges. Theorem 2.5.8 shows that $\alpha > 1/2$ is also necessary for this conclusion. Notice that there is absolute convergence, i.e., $\sum |X_n| < \infty$, if and only if $\alpha > 1$.

Theorem 2.5.6 is sufficient for all of our applications, but our treatment would not be complete if we did not mention the last word on convergence of random series.

Theorem 2.5.8 (Kolmogorov's three-series theorem) Let $X_1, X_2, \dots$ be independent. Let $A > 0$ and let $Y_i = X_i 1_{(|X_i| \leq A)}$. In order that $\sum_{n=1}^{\infty} X_n$ converges a.s., it is necessary and sufficient that

$$(i) \sum_{n=1}^{\infty} P(|X_n| > A) < \infty, (ii) \sum_{n=1}^{\infty} EY_n \text{ converges, and } (iii) \sum_{n=1}^{\infty} \text{var}(Y_n) < \infty$$

Proof We will prove the necessity in Example 3.4.12 as an application of the central limit theorem. To prove the sufficiency, let $\mu_n = EY_n$. (iii) and Theorem 2.5.6 imply that $\sum_{n=1}^{\infty} (Y_n - \mu_n)$ converges a.s. Using (ii) now gives that $\sum_{n=1}^{\infty} Y_n$ converges a.s. (i) and the Borel-Cantelli lemma imply $P(X_n \neq Y_n \text{ i.o.}) = 0$, so $\sum_{n=1}^{\infty} X_n$ converges a.s. $\square$

The link between convergence of series and the strong law of large numbers is provided by

Theorem 2.5.9 (Kronecker's lemma) If $a_n \uparrow \infty$ and $\sum_{n=1}^{\infty} x_n/a_n$ converges, then

$$a_n^{-1} \sum_{m=1}^n x_m \rightarrow 0$$

Proof Let $a_0 = 0$, $b_0 = 0$, and for $m \geq 1$, let $b_m = \sum_{k=1}^m x_k/a_k$. Then $x_m = a_m(b_m - b_{m-1})$ and so

$$\begin{aligned} a_n^{-1} \sum_{m=1}^n x_m &= a_n^{-1} \left\{ \sum_{m=1}^n a_m b_m - \sum_{m=1}^n a_m b_{m-1} \right\} \\ &= a_n^{-1} \left\{ a_n b_n + \sum_{m=2}^n a_{m-1} b_{m-1} - \sum_{m=1}^n a_m b_{m-1} \right\} \\ &= b_n - \sum_{m=1}^n \frac{(a_m - a_{m-1})}{a_n} b_{m-1} \end{aligned}$$

(Recall $a_0 = 0$.) By hypothesis, $b_n \rightarrow b_\infty$ as $n \rightarrow \infty$. Since $a_m - a_{m-1} \geq 0$, the last sum is an average of $b_0, \dots, b_n$. Intuitively, if $\epsilon > 0$ and $M < \infty$ are fixed and n is large, the average assigns mass $\geq 1 - \epsilon$ to the b_m with $m \geq M$, so

$$\sum_{m=1}^n \frac{(a_m - a_{m-1})}{a_n} b_{m-1} \rightarrow b_\infty$$

To argue formally, let $B = \sup |b_n|$, pick M so that $|b_m - b_\infty| < \epsilon/2$ for $m \geq M$, then pick N so that $a_M/a_n < \epsilon/4B$ for $n \geq N$. Now if $n \geq N$, we have

$$\begin{aligned} \left| \sum_{m=1}^n \frac{(a_m - a_{m-1})}{a_n} b_{m-1} - b_\infty \right| &\leq \sum_{m=1}^n \frac{(a_m - a_{m-1})}{a_n} |b_{m-1} - b_\infty| \\ &\leq \frac{a_M}{a_n} \cdot 2B + \frac{a_n - a_M}{a_n} \cdot \frac{\epsilon}{2} < \epsilon \end{aligned}$$

proving the desired result since ϵ is arbitrary. $\square$

Theorem 2.5.10 (The strong law of large numbers) *Let $X_1, X_2, \dots$ be i.i.d. random variables with $E|X_i| < \infty$. Let $EX_i = \mu$ and $S_n = X_1 + \dots + X_n$. Then $S_n/n \rightarrow \mu$ a.s. as $n \rightarrow \infty$.*

Proof Let $Y_k = X_k 1_{(|X_k| \leq k)}$ and $T_n = Y_1 + \dots + Y_n$. By (a) in the proof of Theorem 2.4.1 it suffices to show that $T_n/n \rightarrow \mu$. Let $Z_k = Y_k - EY_k$, so $EZ_k = 0$. Now $\text{var}(Z_k) = \text{var}(Y_k) \leq EY_k^2$ and (b) in the proof of Theorem 2.4.1 imply

$$\sum_{k=1}^{\infty} \text{var}(Z_k)/k^2 \leq \sum_{k=1}^{\infty} EY_k^2/k^2 < \infty$$

Applying Theorem 2.5.6 now, we conclude that $\sum_{k=1}^{\infty} Z_k/k$ converges a.s. so Theorem 2.5.9 implies

$$n^{-1} \sum_{k=1}^n (Y_k - EY_k) \rightarrow 0 \quad \text{and hence} \quad \frac{T_n}{n} - n^{-1} \sum_{k=1}^n EY_k \rightarrow 0 \text{ a.s.}$$

The dominated convergence theorem implies $EY_k \rightarrow \mu$ as $k \rightarrow \infty$. From this, it follows easily that $n^{-1} \sum_{k=1}^n EY_k \rightarrow \mu$, and hence $T_n/n \rightarrow \mu$. $\square$

2.5.1 Rates of Convergence

As mentioned earlier, one of the advantages of the random series proof is that it provides estimates on the rate of convergence of $S_n/n \rightarrow \mu$. By subtracting μ from each random variable, we can and will suppose without loss of generality that $\mu = 0$.

Theorem 2.5.11 *Let $X_1, X_2, \dots$ be i.i.d. random variables with $EX_i = 0$ and $EX_i^2 = \sigma^2 < \infty$. Let $S_n = X_1 + \dots + X_n$. If $\epsilon > 0$ then*

$$S_n/n^{1/2}(\log n)^{1/2+\epsilon} \rightarrow 0 \quad \text{a.s.}$$

Remark The law of the iterated logarithm, Theorem 8.5.2 will show that

$$\limsup_{n \rightarrow \infty} S_n/n^{1/2}(\log \log n)^{1/2} = \sigma\sqrt{2} \quad \text{a.s.}$$

so the last result is not far from the best possible.

Proof Let $a_n = n^{1/2}(\log n)^{1/2+\epsilon}$ for $n \geq 2$ and $a_1 > 0$.

$$\sum_{n=1}^{\infty} \text{var}(X_n/a_n) = \sigma^2 \left(\frac{1}{a_1^2} + \sum_{n=2}^{\infty} \frac{1}{n(\log n)^{1+2\epsilon}} \right) < \infty$$

so applying Theorem 2.5.6, we get $\sum_{n=1}^{\infty} X_n/a_n$ converges a.s. and the indicated result follows from Theorem 2.5.9. $\square$

The next result due to Marcinkiewicz and Zygmund treats the situation in which $EX_i^2 = \infty$ but $E|X_i|^p < \infty$ for some $1 < p < 2$.

Theorem 2.5.12 *Let $X_1, X_2, \dots$ be i.i.d. with $EX_1 = 0$ and $E|X_1|^p < \infty$ where $1 < p < 2$. If $S_n = X_1 + \dots + X_n$, then $S_n/n^{1/p} \rightarrow 0$ a.s.*

Proof Let $Y_k = X_k 1_{(|X_k| \leq k^{1/p})}$ and $T_n = Y_1 + \dots + Y_n$.

$$\sum_{k=1}^{\infty} P(Y_k \neq X_k) = \sum_{k=1}^{\infty} P(|X_k|^p > k) \leq E|X_k|^p < \infty$$

so the Borel-Cantelli lemma implies $P(Y_k \neq X_k \text{ i.o.}) = 0$, and it suffices to show $T_n/n^{1/p} \rightarrow 0$. Using $\text{var}(Y_m) \leq E(Y_m^2)$, Lemma 2.2.13 with $p = 2$, $P(|Y_m| > y) \leq P(|X_1| > y)$, and Fubini's theorem (everything is ≥ 0), we have

$$\begin{aligned} \sum_{m=1}^{\infty} \text{var}(Y_m/m^{1/p}) &\leq \sum_{m=1}^{\infty} EY_m^2/m^{2/p} \\ &\leq \sum_{m=1}^{\infty} \sum_{n=1}^m \int_{(n-1)^{1/p}}^{n^{1/p}} \frac{2y}{m^{2/p}} P(|X_1| > y) dy \\ &= \sum_{n=1}^{\infty} \int_{(n-1)^{1/p}}^{n^{1/p}} \sum_{m=n}^{\infty} \frac{2y}{m^{2/p}} P(|X_1| > y) dy \end{aligned}$$

To bound the integral, we note that for $n \geq 2$ comparing the sum with the integral of $x^{-2/p}$

$$\sum_{m=n}^{\infty} m^{-2/p} \leq \frac{p}{2-p} (n-1)^{(p-2)/p} \leq Cy^{p-2}$$

when $y \in [(n-1)^{1/p}, n^{1/p}]$. Since $E|X_i|^p = \int_0^{\infty} px^{p-1} P(|X_i| > x) dx < \infty$, it follows that

$$\sum_{m=1}^{\infty} \text{var}(Y_m/m^{1/p}) < \infty$$

If we let $\mu_m = EY_m$ and apply Theorems 2.5.6 and 2.5.9, it follows that

$$n^{-1/p} \sum_{m=1}^n (Y_m - \mu_m) \rightarrow 0 \quad \text{a.s.}$$

To estimate μ_m , we note that since $EX_m = 0$, $\mu_m = -E(X_i; |X_i| > m^{1/p})$, so

$$\begin{aligned} |\mu_m| &\leq E(|X|; |X_i| > m^{1/p}) = m^{1/p} E(|X|/m^{1/p}; |X_i| > m^{1/p}) \\ &\leq m^{1/p} E((|X|/m^{1/p})^p; |X_i| > m^{1/p}) \\ &\leq m^{-1+1/p} p^{-1} E(|X_i|^p; |X_i| > m^{1/p}) \end{aligned}$$

Now $\sum_{m=1}^n m^{-1+1/p} \leq Cn^{1/p}$ and $E(|X_i|^p; |X_i| > m^{1/p}) \rightarrow 0$ as $m \rightarrow \infty$, so $n^{-1/p} \sum_{m=1}^n \mu_m \rightarrow 0$ and the desired result follows. $\square$

2.5.2 Infinite Mean

The St. Petersburg game, discussed in Example 2.2.16 and Exercise 2.3.20, is a situation in which $EX_i = \infty$, $S_n/n \log_2 n \rightarrow 1$ in probability but

$$\limsup_{n \rightarrow \infty} S_n/(n \log_2 n) = \infty \text{ a.s.}$$

The next result, due to Feller (1946), shows that when $E|X_1| = \infty$, S_n/a_n cannot converge almost surely to a nonzero limit. In Theorem 2.3.8 we considered the special case $a_n = n$.

Theorem 2.5.13 *Let $X_1, X_2, \dots$ be i.i.d. with $E|X_1| = \infty$ and let $S_n = X_1 + \dots + X_n$. Let a_n be a sequence of positive numbers with a_n/n increasing. Then $\limsup_{n \rightarrow \infty} |S_n|/a_n = 0$ or ∞ according as $\sum_n P(|X_1| \geq a_n) < \infty$ or $= \infty$.*

Proof Since $a_n/n \uparrow$, $a_{kn} \geq ka_n$ for any integer k . Using this and $a_n \uparrow$,

$$\sum_{n=1}^{\infty} P(|X_1| \geq ka_n) \geq \sum_{n=1}^{\infty} P(|X_1| \geq a_{kn}) \geq \frac{1}{k} \sum_{m=k}^{\infty} P(|X_1| \geq a_m)$$

The last observation shows that if the sum is infinite, $\limsup_{n \rightarrow \infty} |X_n|/a_n = \infty$. Since $\max\{|S_{n-1}|, |S_n|\} \geq |X_n|/2$, it follows that $\limsup_{n \rightarrow \infty} |S_n|/a_n = \infty$.

To prove the other half, we begin with the identity

$$(*) \quad \sum_{m=1}^{\infty} mP(a_{m-1} \leq |X_i| < a_m) = \sum_{n=1}^{\infty} P(|X_i| \geq a_{n-1})$$

To see this, write $m = \sum_{n=1}^m 1$ and then use Fubini's theorem. We now let $Y_n = X_n 1_{(|X_n| < a_n)}$, and $T_n = Y_1 + \cdots + Y_n$. When the sum is finite, $P(Y_n \neq X_n \text{ i.o.}) = 0$, and it suffices to investigate the behavior of the T_n . To do this, we let $a_0 = 0$ and compute

$$\begin{aligned} \sum_{n=1}^{\infty} \text{var}(Y_n/a_n) &\leq \sum_{n=1}^{\infty} EY_n^2/a_n^2 \\ &= \sum_{n=1}^{\infty} a_n^{-2} \sum_{m=1}^n \int_{[a_{m-1}, a_m)} y^2 dF(y) \\ &= \sum_{m=1}^{\infty} \int_{[a_{m-1}, a_m)} y^2 dF(y) \sum_{n=m}^{\infty} a_n^{-2} \end{aligned}$$

Since $a_n \geq na_m/m$, we have $\sum_{n=m}^{\infty} a_n^{-2} \leq (m^2/a_m^2) \sum_{n=m}^{\infty} n^{-2} \leq Cma_m^{-2}$, so

$$\leq C \sum_{m=1}^{\infty} m \int_{[a_{m-1}, a_m)} dF(y)$$

Using (*) now, we conclude $\sum_{n=1}^{\infty} \text{var}(Y_n/a_n) < \infty$.

The last step is to show $ET_n/a_n \rightarrow 0$. To begin, we note that if $E|X_i| = \infty$, $\sum_{n=1}^{\infty} P(|X_i| > a_n) < \infty$, and $a_n/n \uparrow$, we must have $a_n/n \uparrow \infty$. To estimate ET_n/a_n now, we observe that

$$\begin{aligned} \left| a_n^{-1} \sum_{m=1}^n EY_m \right| &\leq a_n^{-1} n \sum_{m=1}^n E(|X_m|; |X_m| < a_m) \\ &\leq \frac{na_N}{a_n} + \frac{n}{a_n} E(|X_i|; a_N \leq |X_i| < a_n) \end{aligned}$$

where the last inequality holds for any fixed N . Since $a_n/n \rightarrow \infty$, the first term converges to 0. Since $m/a_m \downarrow$, the second is

$$\begin{aligned} &\leq \sum_{m=N+1}^n \frac{m}{a_m} E(|X_i|; a_{m-1} \leq |X_i| < a_m) \\ &\leq \sum_{m=N+1}^{\infty} m P(a_{m-1} \leq |X_i| < a_m) \end{aligned}$$

(*) shows that the sum is finite, so it is small if N is large and the desired result follows. $\square$

Exercises

- 2.5.1 Suppose $X_1, X_2, \dots$ are i.i.d. with $EX_i = 0$, $\text{var}(X_i) = C < \infty$. Use Theorem 2.5.5 with $n = m^\alpha$ where $\alpha(2p - 1) > 1$ to conclude that if $S_n = X_1 + \cdots + X_n$ and $p > 1/2$, then $S_n/n^p \rightarrow 0$ almost surely.
- 2.5.2 The converse of Theorem 2.5.12 is much easier. Let $p > 0$. If $S_n/n^{1/p} \rightarrow 0$ a.s., then $E|X_1|^p < \infty$.

2.5.3 Let $X_1, X_2, \dots$ be i.i.d. standard normals. Show that for any t

$$\sum_{n=1}^{\infty} X_n \cdot \frac{\sin(n\pi t)}{n} \quad \text{converges a.s.}$$

2.5.4 Let $X_1, X_2, \dots$ be independent with $EX_n = 0$, $\text{var}(X_n) = \sigma_n^2$. (i) Show that if $\sum_n \sigma_n^2/n^2 < \infty$, then $\sum_n X_n/n$ converges a.s., and hence $n^{-1} \sum_{m=1}^n X_m \rightarrow 0$ a.s. (ii) Suppose $\sum \sigma_n^2/n^2 = \infty$ and without loss of generality that $\sigma_n^2 \leq n^2$ for all n . Show that there are independent random variables X_n with $EX_n = 0$ and $\text{var}(X_n) \leq \sigma_n^2$ so that X_n/n , and hence $n^{-1} \sum_{m \leq n} X_m$ does not converge to 0 a.s.

2.5.5 Let $X_n \geq 0$ be independent for $n \geq 1$. The following are equivalent:

- (i) $\sum_{n=1}^{\infty} X_n < \infty$ a.s. (ii) $\sum_{n=1}^{\infty} [P(X_n > 1) + E(X_n 1_{(X_n \leq 1)})] < \infty$
 (iii) $\sum_{n=1}^{\infty} E(X_n/(1 + X_n)) < \infty$.

2.5.6 Let $\psi(x) = x^2$ when $|x| \leq 1$ and $= |x|$ when $|x| \geq 1$. Show that if $X_1, X_2, \dots$ are independent with $EX_n = 0$ and $\sum_{n=1}^{\infty} E\psi(X_n) < \infty$, then $\sum_{n=1}^{\infty} X_n$ converges a.s.

2.5.7 Let X_n be independent. Suppose $\sum_{n=1}^{\infty} E|X_n|^{p(n)} < \infty$, where $0 < p(n) \leq 2$ for all n and $EX_n = 0$ when $p(n) > 1$. Show that $\sum_{n=1}^{\infty} X_n$ converges a.s.

2.5.8 Let $X_1, X_2, \dots$ be i.i.d. and not $\equiv 0$. Then the radius of convergence of the power series $\sum_{n \geq 1} X_n(\omega) z^n$ (i.e., $r(\omega) = \sup\{c : \sum |X_n(\omega)| c^n < \infty\}$) is 1 a.s. or 0 a.s., according as $E \log^+ |X_1| < \infty$ or $= \infty$, where $\log^+ x = \max(\log x, 0)$.

2.5.9 Let $X_1, X_2, \dots$ be independent and let $S_{m,n} = X_{m+1} + \dots + X_n$. Then

$$(\star) \quad P\left(\max_{m < j \leq n} |S_{m,j}| > 2a\right) \min_{m < k \leq n} P(|S_{k,n}| \leq a) \leq P(|S_{m,n}| > a)$$

2.5.10 Use $(\star)$ to prove a theorem of P. Lévy: Let $X_1, X_2, \dots$ be independent and let $S_n = X_1 + \dots + X_n$. If $\lim_{n \rightarrow \infty} S_n$ exists in probability, then it also exists a.s.

2.5.11 Let $X_1, X_2, \dots$ be i.i.d. and $S_n = X_1 + \dots + X_n$. Use $(\star)$ to conclude that if $S_n/n \rightarrow 0$ in probability, then $(\max_{1 \leq m \leq n} S_m)/n \rightarrow 0$ in probability.

2.5.12 Let $X_1, X_2, \dots$ be i.i.d. and $S_n = X_1 + \dots + X_n$. Suppose $a_n \uparrow \infty$ and $a(2^n)/a(2^{n-1})$ is bounded. (i) Use $(\star)$ to show that if $S_n/a(n) \rightarrow 0$ in probability and $S_{2^n}/a(2^n) \rightarrow 0$ a.s., then $S_n/a(n) \rightarrow 0$ a.s. (ii) Suppose in addition that $EX_1 = 0$ and $EX_1^2 < \infty$. Use the previous exercise and Chebyshev's inequality to conclude that $S_n/n^{1/2}(\log_2 n)^{1/2+\epsilon} \rightarrow 0$ a.s.

2.6 Renewal Theory*

Let $\xi_1, \xi_2, \dots$ be i.i.d. positive random variables (i.e., $P(\xi_i > 0) = 1$) with distribution F , and define a sequence of times by $T_0 = 0$, and $T_k = T_{k-1} + \xi_k$ for $k \geq 1$. As explained in Example 2.4.6, we think of ξ_i as the lifetime of the i th light bulb, and T_k is the time the k th bulb burns out. A second interpretation from the discussion of Poisson process in Section 3.6 is that T_k is the time of arrival of the k th customer. A third interpretation from Chapter 5 is that T_k is the time of the k th visit to state k .

To have a neutral terminology, we will refer to the T_k as **renewals**. The term renewal refers to the fact that the process “starts afresh” at T_k , i.e., $\{T_{k+j} - T_k, j \geq 1\}$ has the same distribution as $\{T_j, j \geq 1\}$.



Figure 2.1 Renewal sequence.

Departing slightly from the notation in Example 2.4.6, we let $N_t = \inf\{k : T_k > t\}$. N_t is the number of renewals in $[0, t]$, counting the renewal at time 0. The advantage of this definition is that N_t is a stopping time, i.e., $\{N_t = k\}$ is measurable with respect to $\mathcal{F}_k$.

In Theorem 2.4.7, we showed that:

Theorem 2.6.1 As $t \rightarrow \infty$, $N_t/t \rightarrow 1/\mu$ a.s., where $\mu = E\xi_i \in (0, \infty]$ and $1/\infty = 0$.

Our next result concerns the asymptotic behavior of $U(t) = EN_t$. To derive the result we need:

Theorem 2.6.2 (Wald's equation) Let $X_1, X_2, \dots$ be i.i.d. with $E|X_i| < \infty$. If N is a stopping time with $EN < \infty$, then $ES_N = EX_1EN$.

Proof First suppose the $X_i \geq 0$.

$$ES_N = \int S_N dP = \sum_{n=1}^{\infty} \int S_n 1_{\{N=n\}} dP = \sum_{n=1}^{\infty} \sum_{m=1}^n \int X_m 1_{\{N=n\}} dP$$

Since the $X_i \geq 0$, we can interchange the order of summation (i.e., use Fubini's theorem) to conclude that the last expression

$$= \sum_{m=1}^{\infty} \sum_{n=m}^{\infty} \int X_m 1_{\{N=n\}} dP = \sum_{m=1}^{\infty} \int X_m 1_{\{N \geq m\}} dP$$

Now $\{N \geq m\} = \{N \leq m-1\}^c \in \mathcal{F}_{m-1}$ and is independent of X_m , so the last expression

$$= \sum_{m=1}^{\infty} EX_m P(N \geq m) = EX_1 EN$$

To prove the result in general, we run the last argument backward. If we have $EN < \infty$, then

$$\infty > \sum_{m=1}^{\infty} E|X_m| P(N \geq m) = \sum_{m=1}^{\infty} \sum_{n=m}^{\infty} \int |X_m| 1_{\{N=n\}} dP$$

The last formula shows that the double sum converges absolutely in one order, so Fubini's theorem gives

$$\sum_{m=1}^{\infty} \sum_{n=m}^{\infty} \int X_m 1_{\{N=n\}} dP = \sum_{n=1}^{\infty} \sum_{m=1}^n \int X_m 1_{\{N=n\}} dP$$

Using the independence of $\{N \geq m\} \in \mathcal{F}_{m-1}$ and X_m , and rewriting the last identity, it follows that

$$\sum_{m=1}^{\infty} EX_m P(N \geq m) = ES_N$$

Since the left-hand side is $ENEX_1$, the proof is complete. $\square$

Theorem 2.6.3 As $t \rightarrow \infty$, $U(t)/t \rightarrow 1/\mu$.

Proof We will apply Wald's equation to the stopping time N_t . The first step is to show that $EN_t < \infty$. To do this, pick $\delta > 0$ so that $P(\xi_i > \delta) = \epsilon > 0$ and pick K so that $K\delta \geq t$. Since K consecutive ξ'_i 's that are $> \delta$ will make $T_n > t$, we have

$$P(N_t > mK) \leq (1 - \epsilon^K)^m$$

and $EN_t < \infty$. If $\mu < \infty$, applying Wald's equation now gives

$$\mu EN_t = ET_{N_t} \geq t$$

so $U(t) \geq t/\mu$. The last inequality is trivial when $\mu = \infty$ so it holds in general.

Turning to the upper bound, we observe that if $P(\xi_i \leq c) = 1$, then repeating the last argument shows $\mu EN_t = ES_{N_t} \leq t + c$, and the result holds for bounded distributions. If we let $\bar{\xi}_i = \xi_i \wedge c$ and define $\bar{T}_n$ and $\bar{N}_t$ in the obvious way, then

$$EN_t \leq E\bar{N}_t \leq (t + c)/E(\bar{\xi}_i)$$

Letting $t \rightarrow \infty$ and then $c \rightarrow \infty$ gives $\limsup_{t \rightarrow \infty} EN_t/t \leq 1/\mu$, and the proof is complete. $\square$

To take a closer look at when the renewals occur, we let

$$U(A) = \sum_{n=0}^{\infty} P(T_n \in A)$$

U is called the **renewal measure**. We absorb the old definition, $U(t) = EN_t$, into the new one by regarding $U(t)$ as shorthand for $U([0, t])$. This should not cause problems since $U(t)$ is the distribution function for the renewal measure. The asymptotic behavior of $U(t)$ depends upon whether the distribution F is **arithmetic**, i.e., concentrated on $\{\delta, 2\delta, 3\delta, \dots\}$ for some $\delta > 0$, or **nonarithmetic**, i.e., not arithmetic. We will treat the first case in Chapter 5 as an application of Markov chains, so we will restrict our attention to the second case here.

Theorem 2.6.4 (Blackwell's renewal theorem) If F is nonarithmetic, then

$$U([t, t + h]) \rightarrow h/\mu \quad \text{as } t \rightarrow \infty.$$

We will prove the result in the case $\mu < \infty$ by “coupling” following Lindvall (1977) and Athreya, McDonald, and Ney (1978). To set the stage for the proof, we need a definition and some preliminary computations. If $T_0 \geq 0$ is independent of $\xi_1, \xi_2, \dots$ and has distribution G , then $T_k = T_{k-1} + \xi_k$, $k \geq 1$ defines a **delayed renewal process**, and G is the **delay**

distribution. If we let $N_t = \inf\{k : T_k > t\}$ as before and set $V(t) = EN_t$, then breaking things down according to the value of T_0 gives

$$V(t) = \int_0^t U(t-s) dG(s) \quad (2.6.1)$$

The last integral, and all similar expressions here, is intended to include the contribution of any mass G has at 0. If we let $U(r) = 0$ for $r < 0$, then the last equation can be written as $V = U * G$, where $*$ denotes convolution.

Applying similar reasoning to U gives

$$U(t) = 1 + \int_0^t U(t-s) dF(s) \quad (2.6.2)$$

or, introducing convolution notation,

$$U = 1_{[0,\infty)}(t) + U * F.$$

Convolving each side with G (and recalling $G * U = U * G$) gives

$$V = G * U = G + V * F \quad (2.6.3)$$

We know $U(t) \sim t/\mu$. Our next step is to find a G so that $V(t) = t/\mu$. Plugging what we want into (2.6.3) gives

$$\begin{aligned} t/\mu &= G(t) + \int_0^t \frac{t-y}{\mu} dF(y) \\ \text{so} \quad G(t) &= t/\mu - \int_0^t \frac{t-y}{\mu} dF(y) \end{aligned}$$

The integration-by-parts formula is

$$\int_0^t K(y) dH(y) = H(t)K(t) - H(0)K(0) - \int_0^t H(y) dK(y)$$

If we let $H(y) = (y-t)/\mu$ and $K(y) = 1 - F(y)$, then

$$\frac{1}{\mu} \int_0^t 1 - F(y) dy = \frac{t}{\mu} - \int_0^t \frac{t-y}{\mu} dF(y)$$

so we have

$$G(t) = \frac{1}{\mu} \int_0^t 1 - F(y) dy \quad (2.6.4)$$

It is comforting to note that $\mu = \int_{[0,\infty)} 1 - F(y) dy$, so the last formula defines a probability distribution. When the delay distribution G is the one given in (2.6.4), we call the result the **stationary renewal process**. Something very special happens when $F(t) = 1 - \exp(-\lambda t)$, $t \geq 0$, where $\lambda > 0$ (i.e., the renewal process is a rate λ Poisson process). In this case, $\mu = 1/\lambda$ so $G(t) = F(t)$.

Proof of Theorem 2.6.4 for $\mu < \infty$. Let T_n be a renewal process (with $T_0 = 0$) and T'_n be an independent stationary renewal process. Our first goal is to find J and K so that $|T_J - T'_K| < \epsilon$ and the increments $\{T_{J+i} - T_J, i \geq 1\}$ and $\{T'_{K+i} - T'_K, i \geq 1\}$ are i.i.d. sequences independent of what has come before.

Let $\eta_1, \eta_2, \dots$ and $\eta'_1, \eta'_2, \dots$ be i.i.d. independent of T_n and T'_n and take the values 0 and 1 with probability 1/2 each. Let $v_n = \eta_1 + \dots + \eta_n$ and $v'_n = 1 + \eta'_1 + \dots + \eta'_n$, $S_n = T_{v_n}$ and $S'_n = T'_{v'_n}$. The increments of $S_n - S'_n$ are 0 with probability at least 1/4, and the support of their distribution is symmetric and contains the support of the ξ_k so if the distribution of the ξ_k is nonarithmetic, the random walk $S_n - S'_n$ is irreducible. Since the increments of $S_n - S'_n$ have mean 0, $N = \inf\{n : |S_n - S'_n| < \epsilon\}$ has $P(N < \infty) = 1$, and we can let $J = v_N$ and $K = v'_N$. Let

$$T''_n = \begin{cases} T_n & \text{if } J \geq n \\ T_J + T'_{K+(n-J)} - T'_K & \text{if } J < n \end{cases}$$

In other words, the increments $T''_{J+i} - T''_J$ are the same as $T'_{K+i} - T'_K$ for $i \geq 1$.

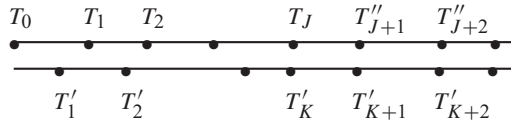


Figure 2.2 Coupling of renewal processes.

It is easy to see from the construction that T_n and T''_n have the same distribution. If we let

$$N'[s, t] = |\{n : T'_n \in [s, t]\}| \quad \text{and} \quad N''[s, t] = |\{n : T''_n \in [s, t]\}|$$

be the number of renewals in $[s, t]$ in the two processes, then on $\{T_J \leq t\}$

$$N''[t, t+h] = N'[t + T'_K - T_J, t+h + T'_K - T_J] \begin{cases} \geq N'[t + \epsilon, t+h - \epsilon] \\ \leq N'[t - \epsilon, t+h + \epsilon] \end{cases}$$

To relate the expected number of renewals in the two processes, we observe that even if we condition on the location of all the renewals in $[0, s]$, the expected number of renewals in $[s, s+t]$ is at most $U(t)$, since the worst thing that could happen is to have a renewal at time s . Combining the last two observations, we see that if $\epsilon < h/2$ (so $[t + \epsilon, t+h - \epsilon]$ has positive length)

$$\begin{aligned} U([t, t+h]) &= EN''[t, t+h] \geq E(N'[t + \epsilon, t+h - \epsilon]; T_J \leq t) \\ &\geq \frac{h - 2\epsilon}{\mu} - P(T_J > t)U(h) \end{aligned}$$

since $EN'[t + \epsilon, t+h - \epsilon] = (h - 2\epsilon)/\mu$ and $\{T_J > t\}$ is determined by the renewals of T in $[0, t]$ and the renewals of T' in $[0, t + \epsilon]$. For the other direction, we observe

$$\begin{aligned} U([t, t+h]) &\leq E(N'[t - \epsilon, t+h + \epsilon]; T_J \leq t) + E(N''[t, t+h]; T_J > t) \\ &\leq \frac{h + 2\epsilon}{\mu} + P(T_J > t)U(h) \end{aligned}$$

The desired result now follows from the fact that $P(T_J > t) \rightarrow 0$ and $\epsilon < h/2$ is arbitrary. $\square$

Proof of Theorem 2.6.4 for $\mu = \infty$. In this case, there is no stationary renewal process, so we have to resort to other methods. Let

$$\beta = \limsup_{t \rightarrow \infty} U(t, t+1] = \lim_{k \rightarrow \infty} U(t_k, t_k+1]$$

for some sequence $t_k \rightarrow \infty$. We want to prove that $\beta = 0$, for then by addition the previous conclusion holds with 1 replaced by any integer n and, by monotonicity, with n replaced by any $h < n$, and this gives us the result in Theorem 2.6.4. Fix i and let

$$a_{k,j} = \int_{(j-1, j]} U(t_k - y, t_k + 1 - y] dF^{i*}(y)$$

By considering the location of T_i we get

$$(a) \quad \lim_{k \rightarrow \infty} \sum_{j=1}^{\infty} a_{k,j} = \lim_{k \rightarrow \infty} \int U(t_k - y, t_k + 1 - y] dF^{i*}(y) = \beta$$

Since β is the lim sup, we must have

$$(b) \quad \limsup_{k \rightarrow \infty} a_{k,j} \leq \beta \cdot P(T_i \in (j-1, j])$$

We want to conclude from (a) and (b) that

$$(c) \quad \liminf_{k \rightarrow \infty} a_{k,j} \geq \beta \cdot P(T_i \in (j-1, j])$$

To do this, we observe that by considering the location of the first renewal in $(j-1, j]$

$$(d) \quad 0 \leq a_{k,j} \leq U(1)P(T_i \in (j-1, j])$$

(c) is trivial when $\beta = 0$ so we can suppose $\beta > 0$. To argue by contradiction, suppose there exist j_0 and $\epsilon > 0$ so that

$$\liminf_{k \rightarrow \infty} a_{k,j_0} \leq \beta \cdot \{P(T_i \in (j_0-1, j_0]) - \epsilon\}$$

Pick $k_n \rightarrow \infty$ so that

$$a_{k_n, j_0} \rightarrow \beta \cdot \{P(T_i \in (j_0-1, j_0]) - \epsilon\}$$

Using (d), we can pick $J \geq j_0$ so that

$$\limsup_{n \rightarrow \infty} \sum_{j=J+1}^{\infty} a_{k_n, j} \leq U(1) \sum_{j=J+1}^{\infty} P(T_i \in (j-1, j]) \leq \beta\epsilon/2$$

Now an easy argument shows

$$\limsup_{n \rightarrow \infty} \sum_{j=1}^J a_{k_n, j} \leq \sum_{j=1}^J \limsup_{n \rightarrow \infty} a_{k_n, j} \leq \beta \left(\sum_{j=1}^J P(T_i \in (j-1, j]) - \epsilon \right)$$

by (b) and our assumption. Adding the last two results shows

$$\limsup_{n \rightarrow \infty} \sum_{j=1}^{\infty} a_{k_n, j} \leq \beta(1 - \epsilon/2)$$

which contradicts (a), and proves (c).

Now, if $j - 1 < y \leq j$, we have

$$U(t_k - y, t_k + 1 - y) \leq U(t_k - j, t_k + 2 - j]$$

so using (c), it follows that for j with $P(T_i \in (j - 1, j]) > 0$, we must have

$$\liminf_{k \rightarrow \infty} U(t_k - j, t_k + 2 - j] \geq \beta$$

Summing over i , we see that the last conclusion is true when $U(j - 1, j] > 0$.

The support of U is closed under addition. (If x is in the support of F^{m*} and y is in the support of F^{n*} , then $x + y$ is in the support of $F^{(m+n)*}$.) We have assumed F is nonarithmetic, so $U(j - 1, j] > 0$ for $j \geq j_0$. Letting $r_k = t_k - j_0$ and considering the location of the last renewal in $[0, r_k]$ and the index of the T_i gives

$$\begin{aligned} 1 &= \sum_{i=0}^{\infty} \int_0^{r_k} (1 - F(r_k - y)) dF^{i*}(y) = \int_0^{r_k} (1 - F(r_k - y)) dU(y) \\ &\geq \sum_{n=1}^{\infty} (1 - F(2n)) U(r_k - 2n, r_k + 2 - 2n] \end{aligned}$$

Since $\liminf_{k \rightarrow \infty} U(r_k - 2n, r_k + 2 - 2n] \geq \beta$ and

$$\sum_{n=0}^{\infty} (1 - F(2n)) \geq \mu/2 = \infty$$

β must be 0, and the proof is complete. $\square$

Remark Following Lindvall (1977), we have based the proof for $\mu = \infty$ on part of Feller's (1961) proof of the discrete renewal theorem (i.e., for arithmetic distributions). See Freedman (1971b) pages 22–25 for an account of Feller's proof. Purists can find a proof that does everything by coupling in Thorisson (1987).

Our next topic is the **renewal equation**: $H = h + H * F$. Two cases we have seen in (2.6.2) and (2.6.3) are:

Example 2.6.5 $h \equiv 1$: $U(t) = 1 + \int_0^t U(t - s) dF(s)$

Example 2.6.6 $h(t) = G(t)$: $V(t) = G(t) + \int_0^t V(t - s) dF(s)$

The last equation is valid for an arbitrary delay distribution. If we let G be the distribution in (2.6.4) and subtract the last two equations, we get:

Example 2.6.7 $H(t) = U(t) - t/\mu$ satisfies the renewal equation with $h(t) = \frac{1}{\mu} \int_t^\infty 1 - F(s) ds$.

Last but not least, we have an example that is a typical application of the renewal equation.

Example 2.6.8 Let $x > 0$ be fixed, and let $H(t) = P(T_{N(t)} - t > x)$. By considering the value of T_1 , we get

$$H(t) = (1 - F(t + x)) + \int_0^t H(t - s) dF(s)$$

The examples here should provide motivation for:

Theorem 2.6.9 *If h is bounded then the function*

$$H(t) = \int_0^t h(t-s) dU(s)$$

is the unique solution of the renewal equation that is bounded on bounded intervals.

Proof Let $U_n(A) = \sum_{m=0}^n P(T_m \in A)$ and

$$H_n(t) = \int_0^t h(t-s) dU_n(s) = \sum_{m=0}^n (h * F^{m*})(t)$$

Here, F^{m*} is the distribution of T_m , and we have extended the definition of h by setting $h(r) = 0$ for $r < 0$. From the last expression, it should be clear that

$$H_{n+1} = h + H_n * F$$

The fact that $U(t) < \infty$ implies $U(t) - U_n(t) \rightarrow 0$. Since h is bounded,

$$|H_n(t) - H(t)| \leq \|h\|_\infty |U(t) - U_n(t)|$$

and $H_n(t) \rightarrow H(t)$ uniformly on bounded intervals. To estimate the convolution, we note that

$$\begin{aligned} |H_n * F(t) - H * F(t)| &\leq \sup_{s \leq t} |H_n(s) - H(s)| \\ &\leq \|h\|_\infty |U(t) - U_n(t)| \end{aligned}$$

since $U - U_n = \sum_{m=n+1}^\infty F^{m*}$ is increasing in t . Letting $n \rightarrow \infty$ in $H_{n+1} = h + H_n * F$, we see that H is a solution of the renewal equation that is bounded on bounded intervals.

To prove uniqueness, we observe that if H_1 and H_2 are two solutions, then $K = H_1 - H_2$ satisfies $K = K * F$. If K is bounded on bounded intervals, iterating gives $K = K * F^{n*} \rightarrow 0$ as $n \rightarrow \infty$, so $H_1 = H_2$. $\square$

The proof of Theorem 2.6.9 is valid when $F(\infty) = P(\xi_i < \infty) < 1$. In this case, we have a **terminating renewal process**. After a geometric number of trials with mean $1/(1 - F(\infty))$, $T_n = \infty$. This “trivial case” has some interesting applications.

Example 2.6.10 (Pedestrian delay) A chicken wants to cross a road (we won’t ask why) on which the traffic is a Poisson process with rate λ . She needs one unit of time with no arrival to safely cross the road. Let $M = \inf\{t \geq 0 : \text{there are no arrivals in } (t, t+1]\}$ be the waiting time until she starts to cross the road. By considering the time of the first arrival, we see that $H(t) = P(M \leq t)$ satisfies

$$H(t) = e^{-\lambda} + \int_0^1 H(t-y) \lambda e^{-\lambda y} dy$$

Comparing with Example 2.6.5 and using Theorem 2.6.9, we see that

$$H(t) = e^{-\lambda} \sum_{n=0}^{\infty} F^{n*}(t)$$

We could have gotten this answer without renewal theory by noting

$$P(M \leq t) = \sum_{n=0}^{\infty} P(T_n \leq t, T_{n+1} = \infty)$$

The last representation allows us to compute the mean of M . Let μ be the mean of the interarrival time given that it is < 1 , and note that the lack of memory property of the exponential distribution implies

$$\mu = \int_0^1 x \lambda e^{-\lambda x} dx = \int_0^{\infty} - \int_1^{\infty} = \frac{1}{\lambda} - \left(1 + \frac{1}{\lambda}\right) e^{-\lambda}$$

Then, by considering the number of renewals in our terminating renewal process,

$$EM = \sum_{n=0}^{\infty} e^{-\lambda} (1 - e^{-\lambda})^n n \mu = (e^{\lambda} - 1) \mu$$

since if X is a geometric with success probability $e^{-\lambda}$ then $EM = \mu E(X - 1)$.

Example 2.6.11 (Cramér's estimates of ruin) Consider an insurance company that collects money at rate c and experiences i.i.d. claims at the arrival times of a Poisson process N_t with rate 1. If its initial capital is x , its wealth at time t is

$$W_x(t) = x + ct - \sum_{m=1}^{Nt} Y_i$$

Here $Y_1, Y_2, \dots$ are i.i.d. with distribution G and mean μ . Let

$$R(x) = P(W_x(t) \geq 0 \text{ for all } t)$$

be the probability of never going bankrupt starting with capital x . By considering the time and size of the first claim:

$$(a) \quad R(x) = \int_0^{\infty} e^{-s} \int_0^{x+cs} R(x+cs-y) dG(y) ds$$

This does not look much like a renewal equation, but with some ingenuity it can be transformed into one. Changing variables $t = x + cs$

$$R(x) e^{-x/c} = \int_x^{\infty} e^{-t/c} \int_0^t R(t-y) dG(y) \frac{dt}{c}$$

Differentiating w.r.t. x and then multiplying by $e^{x/c}$,

$$R'(x) = \frac{1}{c} R(x) - \int_0^x R(x-y) dG(y) \cdot \frac{1}{c}$$

Integrating x from 0 to w

$$(b) \quad R(w) - R(0) = \frac{1}{c} \int_0^w R(x) dx - \frac{1}{c} \int_0^w \int_0^x R(x-y) dG(y) dx$$

Interchanging the order of integration in the double integral, letting

$$S(w) = \int_0^w R(x) dx$$

using $dG = -d(1 - G)$, and then integrating by parts

$$\begin{aligned} -\frac{1}{c} \int_0^w \int_y^w R(x - y) dx dG(y) &= -\frac{1}{c} \int_0^w S(w - y) dG(y) \\ &= \frac{1}{c} \int_0^w S(w - y) d(1 - G)(y) \\ &= \frac{1}{c} \left\{ -S(w) + \int_0^w (1 - G(y)) R(w - y) dy \right\} \end{aligned}$$

Plugging this into (b), we finally have a renewal equation:

$$(c) \quad R(w) = R(0) + \int_0^w R(w - y) \frac{1 - G(y)}{c} dy$$

It took some cleverness to arrive at the last equation, but it is straightforward to analyze. First, we dismiss a trivial case. If $\mu > c$,

$$\frac{1}{t} \left(ct - \sum_{m=1}^{N_t} Y_i \right) \rightarrow c - \mu < 0 \quad \text{a.s.}$$

so $R(x) \equiv 0$. When $\mu < c$,

$$F(x) = \int_0^x \frac{1 - G(y)}{c} dy$$

is a defective probability distribution with $F(\infty) = \mu/c$. Our renewal equation can be written as

$$(d) \quad R = R(0) + R * F$$

so comparing with Example 2.6.5 and using Theorem 2.6.9 tells us $R(w) = R(0)U(w)$. To complete the solution, we have to compute the constant $R(0)$. Letting $w \rightarrow \infty$ and noticing $R(w) \rightarrow 1$, $U(w) \rightarrow (1 - F(\infty))^{-1} = (1 - \mu/c)^{-1}$, we have $R(0) = 1 - \mu/c$.

The basic fact about solutions of the renewal equation (in the nonterminating case) is:

Theorem 2.6.12 (The renewal theorem) *If F is nonarithmetic and h is directly Riemann integrable, then as $t \rightarrow \infty$*

$$H(t) \rightarrow \frac{1}{\mu} \int_0^\infty h(s) ds$$

Intuitively, this holds since Theorem 2.6.9 implies

$$H(t) = \int_0^t h(t - s) dU(s)$$

and Theorem 2.6.4 implies $dU(s) \rightarrow ds/\mu$ as $s \rightarrow \infty$. We will define directly Riemann integrable shortly. We will start doing the proof and then figure out what we need to assume.

Proof Suppose

$$h(s) = \sum_{k=0}^{\infty} a_k 1_{[k\delta, (k+1)\delta)}(s)$$

where $\sum_{k=0}^{\infty} |a_k| < \infty$. Since $U([t, t + \delta]) \leq U([0, \delta]) < \infty$, it follows easily from Theorem 2.6.4 that

$$\int_0^t h(t-s) dU(s) = \sum_{k=0}^{\infty} a_k U((t - (k+1)\delta, t - k\delta]) \rightarrow \frac{1}{\mu} \sum_{k=0}^{\infty} a_k \delta$$

(Pick K so that $\sum_{k \geq K} |a_k| \leq \epsilon/2U([0, \delta])$ and then T so that

$$|a_k| \cdot |U((t - (k+1)\delta, t - k\delta]) - \delta/\mu| \leq \frac{\epsilon}{2K}$$

for $t \geq T$ and $0 \leq k < K$.) If h is an arbitrary function on $[0, \infty)$, we let

$$I^\delta = \sum_{k=0}^{\infty} \delta \sup\{h(x) : x \in [k\delta, (k+1)\delta)\}$$

$$I_\delta = \sum_{k=0}^{\infty} \delta \inf\{h(x) : x \in [k\delta, (k+1)\delta)\}$$

be upper and lower Riemann sums approximating the integral of h over $[0, \infty)$. Comparing h with the obvious upper and lower bounds that are constant on $[k\delta, (k+1)\delta)$ and using the result for the special case,

$$\frac{I_\delta}{\mu} \leq \liminf_{t \rightarrow \infty} \int_0^t h(t-s) dU(s) \leq \limsup_{t \rightarrow \infty} \int_0^t h(t-s) dU(s) \leq \frac{I^\delta}{\mu}$$

If I^δ and I_δ both approach the same finite limit I as $\delta \rightarrow 0$, then h is said to be **directly Riemann integrable**, and it follows that

$$\int_0^t h(t-s) dU(y) \rightarrow I/\mu$$

□

Remark The word “direct” in the name refers to the fact that while the Riemann integral over $[0, \infty)$ is usually defined as the limit of integrals over $[0, a]$, we are approximating the integral over $[0, \infty)$ directly.

In checking the new hypothesis in Theorem 2.6.12, the following result is useful.

Lemma 2.6.13 *If $h(x) \geq 0$ is decreasing with $h(0) < \infty$ and $\int_0^\infty h(x) dx < \infty$, then h is directly Riemann integrable.*

Proof Because h is decreasing, $I^\delta = \sum_{k=0}^{\infty} \delta h(k\delta)$ and $I_\delta = \sum_{k=0}^{\infty} \delta h((k+1)\delta)$. So

$$I^\delta \geq \int_0^\infty h(x) dx \geq I_\delta = I^\delta - h(0)\delta$$

proving the desired result. □

Returning now to our examples, we skip the first two because, in those cases, $h(t) \rightarrow 1$ as $t \rightarrow \infty$, so h is not integrable in any sense.

Example 2.6.14 (Continuation of Example 2.6.7) $h(t) = \frac{1}{\mu} \int_{[t, \infty)} 1 - F(s) ds$. h is decreasing, $h(0) = 1$, and

$$\begin{aligned} \mu \int_0^\infty h(t) dt &= \int_0^\infty \int_t^\infty 1 - F(s) ds dt \\ &= \int_0^\infty \int_0^s 1 - F(s) dt ds = \int_0^\infty s(1 - F(s)) ds = E(\xi_i^2/2) \end{aligned}$$

So, if $\nu \equiv E(\xi_i^2) < \infty$, it follows from Lemma 2.6.13, Theorem 2.6.12, and the formula in Example 2.6.7 that

$$0 \leq U(t) - t/\mu \rightarrow \nu/2\mu^2 \quad \text{as } t \rightarrow \infty$$

When the renewal process is a rate λ Poisson process, i.e., $P(\xi_i > t) = e^{-\lambda t}$, $N(t) - 1$ has a Poisson distribution with mean λt , so $U(t) = 1 + \lambda t$. According to Feller, Vol. II (1971), p. 385, if the ξ_i are uniform on $(0, 1)$, then

$$U(t) = \sum_{k=0}^n (-1)^k e^{t-k} (t-k)^k / k! \quad \text{for } n \leq t \leq n+1$$

As he says, the exact expression “reveals little about the nature of U . The asymptotic formula $0 \leq U(t) - 2t \rightarrow 2/3$ is much more interesting.”

Example 2.6.15 (Continuation of Example 2.6.8) $h(t) = 1 - F(t+x)$. Again, h is decreasing, but this time $h(0) \leq 1$ and the integral of h is finite when $\mu = E(\xi_i) < \infty$. Applying Lemma 2.6.13 and Theorem 2.6.12 now gives

$$P(T_{N(t)} - t > x) \rightarrow \frac{1}{\mu} \int_0^\infty h(s) ds = \frac{1}{\mu} \int_x^\infty 1 - F(t) dt$$

so (when $\mu < \infty$) the distribution of the **residual waiting time** $T_{N(t)} - t$ converges to the delay distribution that produces the stationary renewal process. This fact also follows from our proof of 2.6.4.

Exercises

2.6.1 Show that $t/E(\xi_i \wedge t) \leq U(t) \leq 2t/E(\xi_i \wedge t)$.

2.6.2 Deduce Theorem 2.6.3 from Theorem 2.6.1 by showing

$$\limsup_{t \rightarrow \infty} E(N_t/t)^2 < \infty.$$

Hint: Use a comparison like the one in the proof of Theorem 2.6.3.

2.6.3 Customers arrive at times of a Poisson process with rate 1. If the server is occupied, they leave. (Think of a public telephone or prostitute.) If not, they enter service and require a service time with a distribution F that has mean μ . Show that the times at which customers enter service are a renewal process with mean $\mu + 1$, and

use Theorem 2.6.1 to conclude that the asymptotic fraction of customers served is $1/(\mu + 1)$.

In the remaining problems we assume that F is nonarithmetic, and in problems where the mean appears we assume it is finite.

- 2.6.4 Let $A_t = t - T_{N(t)-1}$ be the “age” at time t , i.e., the amount of time since the last renewal. If we fix $x > 0$, then $H(t) = P(A_t > x)$ satisfies the renewal equation

$$H(t) = (1 - F(t)) \cdot 1_{(x, \infty)}(t) + \int_0^t H(t-s) dF(s)$$

so $P(A_t > x) \rightarrow \frac{1}{\mu} \int_{(x, \infty)} (1 - F(t)) dt$, which is the limit distribution for the residual lifetime $B_t = T_{N(t)} - t$.

- 2.6.5 Use the renewal equation in the last problem and Theorem 2.6.9 to conclude that if T is a rate λ Poisson process, A_t has the same distribution as $\xi_i \wedge t$.

- 2.6.6 Let $A_t = t - T_{N(t)-1}$ and $B_t = T_{N(t)} - t$. Show that

$$P(A_t > x, B_t > y) \rightarrow \frac{1}{\mu} \int_{x+y}^{\infty} (1 - F(t)) dt$$

- 2.6.7 **Alternating renewal process.** Let $\xi_1, \xi_2, \dots > 0$ be i.i.d. with distribution F_1 and let $\eta_1, \eta_2, \dots > 0$ be i.i.d. with distribution F_2 . Let $T_0 = 0$ and for $k \geq 1$ let $S_k = T_{k-1} + \xi_k$ and $T_k = S_k + \eta_k$. In words, we have a machine that works for an amount of time ξ_k , breaks down, and then requires η_k units of time to be repaired. Let $F = F_1 * F_2$ and let $H(t)$ be the probability the machine is working at time t . Show that if F is nonarithmetic, then as $t \rightarrow \infty$

$$H(t) \rightarrow \mu_1/(\mu_1 + \mu_2)$$

where μ_i is the mean of F_i .

- 2.6.8 Write a renewal equation for $H(t) = P(\text{number of renewals in } [0, t] \text{ is odd})$ and use the renewal theorem to show that $H(t) \rightarrow 1/2$. Note: This is a special case of the previous exercise.

- 2.6.9 **Renewal densities.** Show that if $F(t)$ has a directly Riemann integrable density function $f(t)$, then the $V = U - 1_{[0, \infty)}$ has a density v that satisfies

$$v(t) = f(t) + \int_0^t v(t-s) dF(s)$$

Use the renewal theorem to conclude that if f is directly Riemann integrable, then $v(t) \rightarrow 1/\mu$ as $t \rightarrow \infty$.

2.7 Large Deviations*

Let $X_1, X_2, \dots$ be i.i.d. and let $S_n = X_1 + \dots + X_n$. In this section, we will investigate the rate at which $P(S_n > na) \rightarrow 0$ for $a > \mu = EX_i$. We will ultimately conclude that if the

moment-generating function $\varphi(\theta) = E \exp(\theta X_i) < \infty$ for some $\theta > 0$, $P(S_n \geq na) \rightarrow 0$ exponentially rapidly and we will identify

$$\gamma(a) = \lim_{n \rightarrow \infty} \frac{1}{n} \log P(S_n \geq na)$$

Our first step is to prove that the limit exists. This is based on an observation that will be useful several times later. Let $\pi_n = P(S_n \geq na)$.

$$\pi_{m+n} \geq P(S_m \geq ma, S_{n+m} - S_m \geq na) = \pi_m \pi_n$$

since S_m and $S_{n+m} - S_m$ are independent. Letting $\gamma_n = \log \pi_n$ transforms multiplication into addition.

Lemma 2.7.1 *If $\gamma_{m+n} \geq \gamma_m + \gamma_n$, then as $n \rightarrow \infty$, $\gamma_n/n \rightarrow \sup_m \gamma_m/m$.*

Proof Clearly, $\limsup \gamma_n/n \leq \sup \gamma_m/m$. To complete the proof, it suffices to prove that for any m $\liminf \gamma_n/n \geq \gamma_m/m$. Writing $n = km + \ell$ with $0 \leq \ell < m$ and making repeated use of the hypothesis gives $\gamma_n \geq k\gamma_m + \gamma_\ell$. Dividing by $n = km + \ell$ gives

$$\frac{\gamma(n)}{n} \geq \left(\frac{km}{km + \ell} \right) \frac{\gamma(m)}{m} + \frac{\gamma(\ell)}{n}$$

Letting $n \rightarrow \infty$ and recalling $n = km + \ell$ with $0 \leq \ell < m$ gives the desired result. $\square$

Lemma 2.7.1 implies that $\lim_{n \rightarrow \infty} \frac{1}{n} \log P(S_n \geq na) = \gamma(a)$ exists ≤ 0 . It follows from the formula for the limit that

$$P(S_n \geq na) \leq e^{n\gamma(a)} \quad (2.7.1)$$

The last conclusion is valid for any distribution but it is not very useful if $\gamma(a) = 0$. For the rest of this section, we will suppose:

(H1) $\varphi(\theta) = E \exp(\theta X_i) < \infty$ for some $\theta > 0$

Let $\theta_+ = \sup\{\theta : \varphi(\theta) < \infty\}$, $\theta_- = \inf\{\theta : \varphi(\theta) < \infty\}$ and note that $\varphi(\theta) < \infty$ for $\theta \in (\theta_-, \theta_+)$. (H1) implies that $EX_i^+ < \infty$ so $\mu = EX^+ - EX^- \in [-\infty, \infty)$. If $\theta > 0$ Chebyshev's inequality implies

$$e^{\theta na} P(S_n \geq na) \leq E \exp(\theta S_n) = \varphi(\theta)^n$$

or letting $\kappa(\theta) = \log \varphi(\theta)$

$$P(S_n \geq na) \leq \exp(-n\{a\theta - \kappa(\theta)\}) \quad (2.7.2)$$

Our first goal is to show:

Lemma 2.7.2 *If $a > \mu$ and $\theta > 0$ is small, then $a\theta - \kappa(\theta) > 0$.*

Proof $\kappa(0) = \log \varphi(0) = 0$, so it suffices to show that (i) κ is continuous at 0, (ii) differentiable on $(0, \theta_+)$, and (iii) $\kappa'(\theta) \rightarrow \mu$ as $\theta \rightarrow 0$. For then

$$a\theta - \kappa(\theta) = \int_0^\theta a - \kappa'(x) dx > 0$$

for small θ .

Let $F(x) = P(X_i \leq x)$. To prove (i) we note that if $0 < \theta < \theta_0 < \theta_-$

$$e^{\theta x} \leq 1 + e^{\theta_0 x} \quad (*)$$

so by the dominated convergence theorem as $\theta \rightarrow 0$

$$\int e^{\theta x} dF \rightarrow \int 1 dF = 1$$

To prove (ii) we note that if $|h| < h_0$, then

$$|e^{hx} - 1| = \left| \int_0^{hx} e^y dy \right| \leq |hx| e^{h_0 x}$$

so an application of the dominated convergence theorem shows that

$$\begin{aligned} \varphi'(\theta) &= \lim_{h \rightarrow 0} \frac{\varphi(\theta + h) - \varphi(\theta)}{h} \\ &= \lim_{h \rightarrow 0} \int \frac{e^{hx} - 1}{h} e^{\theta x} dF(x) \\ &= \int x e^{\theta x} dF(x) \quad \text{for } \theta \in (0, \theta_+) \end{aligned}$$

From the last equation, it follows that $\kappa(\theta) = \log \phi(\theta)$ has $\kappa'(\theta) = \phi'(\theta)/\phi(\theta)$. Using (*) and the dominated convergence theorem gives (iii) and the proof is complete. $\square$

Having found an upper bound on $P(S_n \geq na)$, it is natural to optimize it by finding the maximum of $\theta a - \kappa(\theta)$:

$$\frac{d}{d\theta} \{\theta a - \log \varphi(\theta)\} = a - \varphi'(\theta)/\varphi(\theta)$$

so (assuming things are nice) the maximum occurs when $a = \varphi'(\theta)/\varphi(\theta)$. To turn the parenthetical clause into a mathematical hypothesis we begin by defining

$$F_\theta(x) = \frac{1}{\varphi(\theta)} \int_{-\infty}^x e^{\theta y} dF(y)$$

whenever $\phi(\theta) < \infty$. It follows from the proof of Lemma 2.7.2 that if $\theta \in (\theta_-, \theta_+)$, F_θ is a distribution function with mean

$$\int x dF_\theta(x) = \frac{1}{\varphi(\theta)} \int_{-\infty}^{\infty} x e^{\theta x} dF(x) = \frac{\varphi'(\theta)}{\varphi(\theta)}$$

Repeating the proof in Lemma 2.7.2, it is easy to see that if $\theta \in (\theta_-, \theta_+)$, then

$$\phi''(\theta) = \int_{-\infty}^{\infty} x^2 e^{\theta x} dF(x)$$

So we have

$$\frac{d}{d\theta} \frac{\varphi'(\theta)}{\varphi(\theta)} = \frac{\varphi''(\theta)}{\varphi(\theta)} - \left(\frac{\varphi'(\theta)}{\varphi(\theta)} \right)^2 = \int x^2 dF_\theta(x) - \left(\int x dF_\theta(x) \right)^2 \geq 0$$

since the last expression is the variance of F_θ . If we assume

(H2) the distribution F is not a point mass at μ

then $\varphi'(\theta)/\varphi(\theta)$ is strictly increasing and $a\theta - \log \phi(\theta)$ is concave. Since we have $\varphi'(0)/\varphi(0) = \mu$, this shows that for each $a > \mu$ there is at most one $\theta_a \geq 0$ that solves $a = \varphi'(\theta_a)/\varphi(\theta_a)$, and this value of θ maximizes $a\theta - \log \varphi(\theta)$. Before discussing the existence of θ_a , we will consider some examples.

Example 2.7.3 (Normal distribution)

$$\int e^{\theta x} (2\pi)^{-1/2} \exp(-x^2/2) dx = \exp(\theta^2/2) \int (2\pi)^{-1/2} \exp(-(x - \theta)^2/2) dx$$

The integrand in the last integral is the density of a normal distribution with mean θ and variance 1, so $\varphi(\theta) = \exp(\theta^2/2)$, $\theta \in (-\infty, \infty)$. In this case, $\varphi'(\theta)/\varphi(\theta) = \theta$ and

$$F_\theta(x) = e^{-\theta^2/2} \int_{-\infty}^x e^{\theta y} (2\pi)^{-1/2} e^{-y^2/2} dy$$

is a normal distribution with mean θ and variance 1.

Example 2.7.4 (Exponential distribution with parameter λ) If $\theta < \lambda$

$$\int_0^\infty e^{\theta x} \lambda e^{-\lambda x} dx = \lambda/(\lambda - \theta)$$

$\varphi'(\theta)/\varphi(\theta) = 1/(\lambda - \theta)$ and

$$F_\theta(x) = \frac{\lambda}{\lambda - \theta} \int_0^x e^{\theta y} \lambda e^{-\lambda y} dy$$

is an exponential distribution with parameter $\lambda - \theta$, and hence mean $1/(\lambda - \theta)$.

Example 2.7.5 (Coin flips) $P(X_i = 1) = P(X_i = -1) = 1/2$

$$\varphi(\theta) = (e^\theta + e^{-\theta})/2$$

$$\varphi'(\theta)/\varphi(\theta) = (e^\theta - e^{-\theta})/(e^\theta + e^{-\theta})$$

$F_\theta(\{x\})/F(\{x\}) = e^{\theta x}/\varphi(\theta)$ so

$$F_\theta(\{1\}) = e^\theta/(e^\theta + e^{-\theta}) \quad \text{and} \quad F_\theta(\{-1\}) = e^{-\theta}/(e^\theta + e^{-\theta})$$

Example 2.7.6 (Perverted exponential) Let $g(x) = Cx^{-3}e^{-x}$ for $x \geq 1$, $g(x) = 0$ otherwise, and choose C so that g is a probability density. In this case,

$$\varphi(\theta) = \int e^{\theta x} g(x) dx < \infty$$

if and only if $\theta \leq 1$, and when $\theta \leq 1$, we have

$$\frac{\varphi'(\theta)}{\varphi(\theta)} \leq \frac{\varphi'(1)}{\varphi(1)} = \int_1^\infty Cx^{-2} dx / \int_1^\infty Cx^{-3} dx = 2$$

Recall $\theta_+ = \sup\{\theta : \varphi(\theta) < \infty\}$. In Examples 2.7.3 and 2.7.4, we have $\varphi'(\theta)/\varphi(\theta) \uparrow \infty$ as $\theta \uparrow \theta_+$ so we can solve $a = \varphi'(\theta)/\varphi(\theta)$ for any $a > \mu$. In Example 2.7.5, $\varphi'(\theta)/\varphi(\theta) \uparrow 1$ as $\theta \rightarrow \infty$, but we cannot hope for much more since F , and hence F_θ is supported on $\{-1, 1\}$. Example 2.7.6 presents a problem since we cannot solve $a = \varphi'(\theta)/\varphi(\theta)$ when

$a > 2$. Theorem 2.7.10 will cover this problem case, but first we will treat the cases in which we can solve the equation.

Theorem 2.7.7 *Suppose in addition to (H1) and (H2) that there is a $\theta_a \in (0, \theta_+)$ so that $a = \phi'(\theta_a)/\phi(\theta_a)$. Then, as $n \rightarrow \infty$,*

$$n^{-1} \log P(S_n \geq na) \rightarrow -a\theta_a + \log \phi(\theta_a)$$

Proof The fact that the limsup of the left-hand side $\leq$ the right-hand side follows from (2.7.2). To prove the other inequality, pick $\lambda \in (\theta_a, \theta_+)$, let $X_1^\lambda, X_2^\lambda, \dots$ be i.i.d. with distribution F_λ and let $S_n^\lambda = X_1^\lambda + \dots + X_n^\lambda$. Writing dF/dF_λ for the Radon-Nikodym derivative of the associated measures, it is immediate from the definition that $dF/dF_\lambda = e^{-\lambda x} \phi(\lambda)$. If we let F_λ^n and F^n denote the distributions of S_n^λ and S_n , then

Lemma 2.7.8 $\frac{dF^n}{dF_\lambda^n} = e^{-\lambda x} \phi(\lambda)^n.$

Proof We will prove this by induction. The result holds when $n = 1$. For $n > 1$, we note that

$$\begin{aligned} F^n &= F^{n-1} * F(z) = \int_{-\infty}^{\infty} dF^{n-1}(x) \int_{-\infty}^{z-x} dF(y) \\ &= \int dF_\lambda^{n-1}(x) \int dF_\lambda(y) 1_{(x+y \leq z)} e^{-\lambda(x+y)} \phi(\lambda)^n \\ &= E \left(1_{(S_{n-1}^\lambda + X_n^\lambda \leq z)} e^{-\lambda(S_{n-1}^\lambda + X_n^\lambda)} \phi(\lambda)^n \right) \\ &= \int_{-\infty}^z dF_\lambda^n(u) e^{-\lambda u} \phi(\lambda)^n \end{aligned}$$

where in the last two equalities we have used Theorem 1.6.9 for $(S_{n-1}^\lambda, X_n^\lambda)$ and S_n^λ . $\square$

If $\nu > a$, then the lemma and monotonicity imply

$$(*) \quad P(S_n \geq na) \geq \int_{na}^{n\nu} e^{-\lambda x} \phi(\lambda)^n dF_\lambda^n(x) \geq \phi(\lambda)^n e^{-\lambda n\nu} (F_\lambda^n(n\nu) - F_\lambda^n(na))$$

F_λ has mean $\phi'(\lambda)/\phi(\lambda)$, so if we have $a < \phi'(\lambda)/\phi(\lambda) < \nu$, then the weak law of large numbers implies

$$F_\lambda^n(n\nu) - F_\lambda^n(na) \rightarrow 1 \text{ as } n \rightarrow \infty$$

From the last conclusion and (*) it follows that

$$\liminf_{n \rightarrow \infty} n^{-1} \log P(S_n > na) \geq -\lambda\nu + \log \phi(\lambda)$$

Since $\lambda > \theta_a$ and $\nu > a$ are arbitrary, the proof is complete. $\square$

To get a feel for what the answers look like, we consider our examples. To prepare for the computations, we recall some important information:

$$\begin{aligned} \kappa(\theta) &= \log \phi(\theta) & \kappa'(\theta) &= \phi'(\theta)/\phi(\theta) & \theta_a &\text{ solves } \kappa'(\theta_a) = a \\ \gamma(a) &= \lim_{n \rightarrow \infty} (1/n) \log P(S_n \geq na) = -a\theta_a + \kappa(\theta_a) \end{aligned}$$

Normal distribution (Example 2.7.3)

$$\begin{aligned}\kappa(\theta) &= \theta^2/2 & \kappa'(\theta) &= \theta & \theta_a &= a \\ \gamma(a) &= -a\theta_a + \kappa(\theta_a) = -a^2/2\end{aligned}$$

One can check the last result by observing that S_n has a normal distribution with mean 0 and variance n , and then using Theorem 1.2.6.

Exponential distribution (Example 2.7.4) with $\lambda = 1$

$$\begin{aligned}\kappa(\theta) &= -\log(1 - \theta) & \kappa'(\theta) &= 1/(1 - \theta) & \theta_a &= 1 - 1/a \\ \gamma(a) &= -a\theta_a + \kappa(\theta_a) = -a + 1 + \log a\end{aligned}$$

With these two examples as models, the reader should be able to do

Coin flips (Example 2.7.5). Here we take a different approach. To find the θ that makes the mean of $F_\theta = a$, we set $F_\theta(\{1\}) = e^\theta / (e^\theta + e^{-\theta}) = (1 + a)/2$. Letting $x = e^\theta$ gives

$$2x = (1 + a)(x + x^{-1}) \quad (a - 1)x^2 + (1 + a) = 0$$

So $x = \sqrt{(1 + a)/(1 - a)}$ and $\theta_a = \log x = \{\log(1 + a) - \log(1 - a)\}/2$.

$$\begin{aligned}\phi(\theta_a) &= \frac{e^{\theta_a} + e^{-\theta_a}}{2} = \frac{e^{\theta_a}}{1 + a} = \frac{1}{\sqrt{(1 + a)(1 - a)}} \\ \gamma(a) &= -a\theta_a + \kappa(\theta_a) = -\{(1 + a)\log(1 + a) + (1 - a)\log(1 - a)\}/2\end{aligned}$$

In Exercise 3.1.3, this result will be proved by a direct computation. Since the formula for $\gamma(a)$ is rather ugly, the simpler bound in Exercise 2.7.4 is useful.

Turning now to the problematic values for which we cannot solve $a = \phi'(\theta_a)/\phi(\theta_a)$:

Theorem 2.7.9 Suppose $x_o = \sup\{x : F(x) < 1\} < \infty$ and F is not a point mass at x_o . $\phi(\theta) < \infty$ for all $\theta > 0$ and $\phi'(\theta)/\phi(\theta) \rightarrow x_o$ as $\theta \uparrow \infty$.

Proof Since $P(X \leq x_o) = 1$, $Ee^{\theta X} < \infty$ for all $\theta > 0$. Since F_θ is concentrated on $(-\infty, x_o]$ it is clear that $\mu_\theta = \phi'(\theta)/\phi(\theta) \leq x_o$. On the other hand, if $\delta > 0$, then $P(X \geq x_o - \delta) = c_\delta > 0$, $Ee^{\theta X} \geq c_\delta e^{\theta(x_o - \delta)}$, and hence

$$F_\theta(x_o - 2\delta) = \frac{1}{\phi(\theta)} \int_{-\infty}^{x_o - 2\delta} e^{\theta x} dF(x) \leq \frac{e^{x_o - 2\delta}\theta}{c_\delta e^{(x_o - \delta)\theta}} = e^{-\theta\delta}/c_\delta \rightarrow 0 \quad (2.7.3)$$

Since $\delta > 0$ is arbitrary, it follows that $\mu_\theta \rightarrow x_o$. □

The result for $a = x_o$ is trivial:

$$\frac{1}{n} \log P(S_n \geq nx_o) = \log P(X_i = x_o) \quad \text{for all } n$$

We leave it to the reader to show that as $a \uparrow x_o$, $\gamma(a) \downarrow \log P(X_i = x_o)$.

When $x_o = \infty$ and $\theta_+ = \infty$, the computation in (2.7.3) implies $\phi'(\theta)/\phi(\theta) \uparrow \infty$ as $\theta \uparrow \infty$, so the only case that remains is covered by:

Theorem 2.7.10 Suppose $x_0 = \infty$, $\theta_+ < \infty$, and $\varphi'(\theta)/\varphi(\theta)$ increases to a finite limit a_0 as $\theta \uparrow \theta_+$. If $a_0 \leq a < \infty$

$$n^{-1} \log P(S_n \geq na) \rightarrow -a\theta_+ + \log \varphi(\theta_+)$$

i.e., $\gamma(a)$ is linear for $a \geq a_0$.

Proof Since $(\log \varphi(\theta))' = \varphi'(\theta)/\varphi(\theta)$, integrating from 0 to θ_+ shows that $\log(\varphi(\theta_+)) < \infty$. Letting $\theta = \theta_+$ in (2.7.2) shows that the limsup of the left-hand side $\leq$ the right-hand side. To get the other direction, we will use the transformed distribution F_λ , for $\lambda = \theta_+$. Letting $\theta \uparrow \theta_+$ and using the dominated convergence theorem for $x \leq 0$ and the monotone convergence theorem for $x \geq 0$, we see that F_λ has mean a_0 . From (*) in the proof of Theorem 2.7.7, we see that if $a_0 \leq a < v = a + 3\epsilon$

$$P(S_n \geq na) \geq \varphi(\lambda)^n e^{-n\lambda v} (F_\lambda^n(nv) - F_\lambda^n(na))$$

and hence

$$\frac{1}{n} \log P(S_n \geq na) \geq \log \varphi(\lambda) - \lambda v + \frac{1}{n} \log P(S_n^\lambda \in (na, nv])$$

Letting $X_1^\lambda, X_2^\lambda, \dots$ be i.i.d. with distribution F_λ and $S_n^\lambda = X_1^\lambda + \dots + X_n^\lambda$, we have

$$\begin{aligned} P(S_n^\lambda \in (na, nv]) &\geq P\{S_{n-1}^\lambda \in ((a_0 - \epsilon)n, (a_0 + \epsilon)n]\} \\ &\quad \cdot P\{X_n^\lambda \in ((a - a_0 + \epsilon)n, (a - a_0 + 2\epsilon)n]\} \\ &\geq \frac{1}{2} P\{X_n^\lambda \in ((a - a_0 + \epsilon)n, (a - a_0 + \epsilon)(n+1)]\} \end{aligned}$$

for large n by the weak law of large numbers. To get a lower bound on the right-hand side of the last equation, we observe that

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log P(X_1^\lambda \in ((a - a_0 + \epsilon)n, (a - a_0 + \epsilon)(n+1)]) = 0$$

for if the lim sup was < 0 , we would have $E \exp(\eta X_1^\lambda) < \infty$ for some $\eta > 0$ and hence $E \exp((\lambda + \eta)X_1) < \infty$, contradicting the definition of $\lambda = \theta_+$. To finish the argument now, we recall that Theorem 2.7.1 implies that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log P(S_n \geq na) = \gamma(a)$$

exists, so our lower bound on the lim sup is good enough. $\square$

By adapting the proof of the last result, you can show that (H1) is necessary for exponential convergence:

Exercises

- 2.7.1 Consider $\gamma(a)$ defined in (2.7.1). The following are equivalent: (a) $\gamma(a) = -\infty$, (b) $P(X_1 \geq a) = 0$, and (c) $P(S_n \geq na) = 0$ for all n .

- 2.7.2 Use the definition to conclude that if $\lambda \in [0, 1]$ is rational, then $\gamma(\lambda a + (1 - \lambda)b) \geq \lambda\gamma(a) + (1 - \lambda)\gamma(b)$. Use monotonicity to conclude that the last relationship holds for all $\lambda \in [0, 1]$ so γ is concave, and hence Lipschitz continuous on compact subsets of $\gamma(a) > -\infty$.
- 2.7.3 Let $X_1, X_2, \dots$ be i.i.d. Poisson with mean 1, and let $S_n = X_1 + \dots + X_n$. Find $\lim_{n \rightarrow \infty} (1/n) \log P(S_n \geq na)$ for $a > 1$. The answer and another proof can be found in Exercise 3.1.4.
- 2.7.4 Show that for coin flips $\varphi(\theta) \leq \exp(\varphi(\theta) - 1) \leq \exp(\beta\theta^2)$ for $\theta \leq 1$, where $\beta = \sum_{n=1}^{\infty} 1/(2n)! \approx 0.586$, and use (2.7.2) to conclude that $P(S_n \geq an) \leq \exp(-na^2/4\beta)$ for all $a \in [0, 1]$. It is customary to simplify this further by using $\beta \leq \sum_{n=1}^{\infty} 2^{-n} = 1$.
- 2.7.5 Suppose $EX_i = 0$ and $E \exp(\theta X_i) = \infty$ for all $\theta > 0$. Then

$$\frac{1}{n} \log P(S_n \geq na) \rightarrow 0 \text{ for all } a > 0$$

- 2.7.6 Suppose $EX_i = 0$. Show that if $\epsilon > 0$, then

$$\liminf_{n \rightarrow \infty} P(S_n \geq na)/nP(X_1 \geq n(a + \epsilon)) \geq 1$$

Hint: Let $F_n = \{X_i \geq n(a + \epsilon) \text{ for exactly one } i \leq n\}$.